

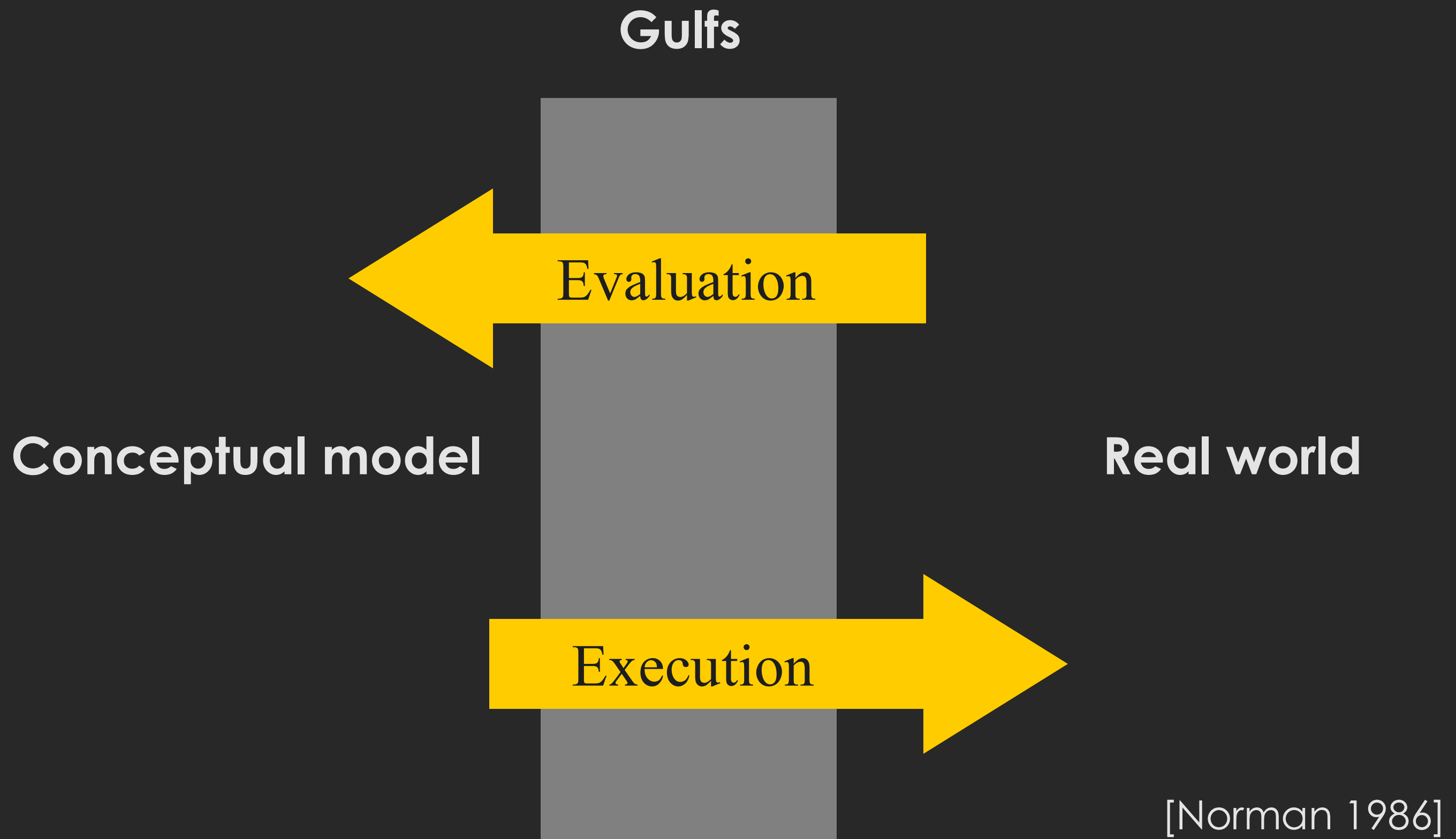
Interaction and Animation

Alex Kale

Slides credited to

Jessica Hullman

Gulfs of execution & evaluation



Gulfs of execution & evaluation

Gulf of Execution

The difference between the user's intentions and the allowable actions.

Gulf of Evaluation

The amount of effort that the person must exert to interpret the state of the system and to determine how well the expectations and intentions have been met.

Interaction techniques

Pointing/selection

Brushing and linking

Dynamic queries

Generalized selections

Pointing/Selection

Select

Precedes operating on a point

- n Get details on demand
- n Filter

Mark something as interesting

- n Mark items to keep track of them
- n Often precedes other operations

Example:

- n Select a mark (e.g. country) in a scatterplot
- n Selecting a placemark in Google Maps

Pointing Methods

Point Selection

Mouse Hover / Click

Touch / Tap

Select Nearby Element (e.g., Bubble Cursor)

Region Selection

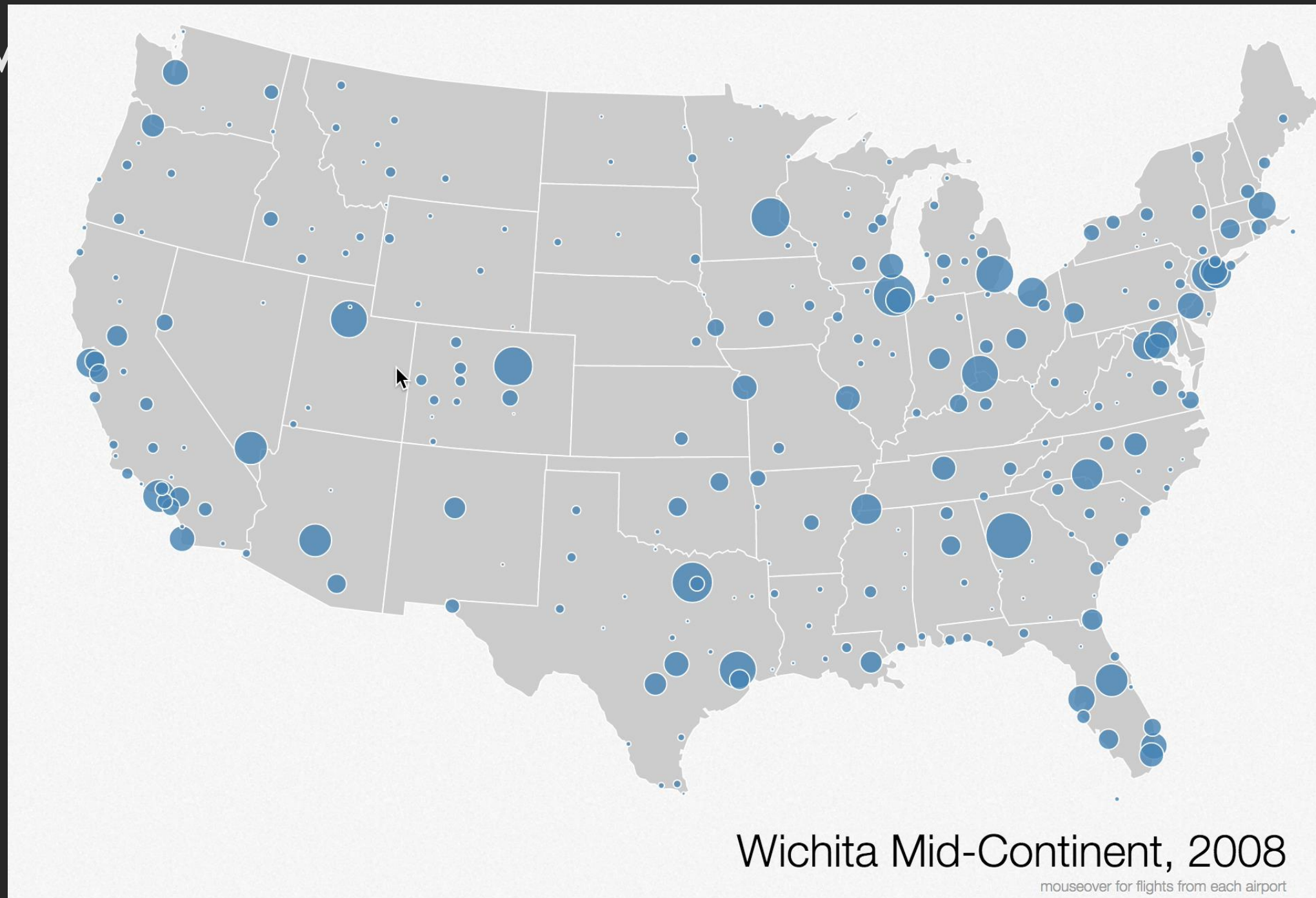
Rubber-band or Lasso

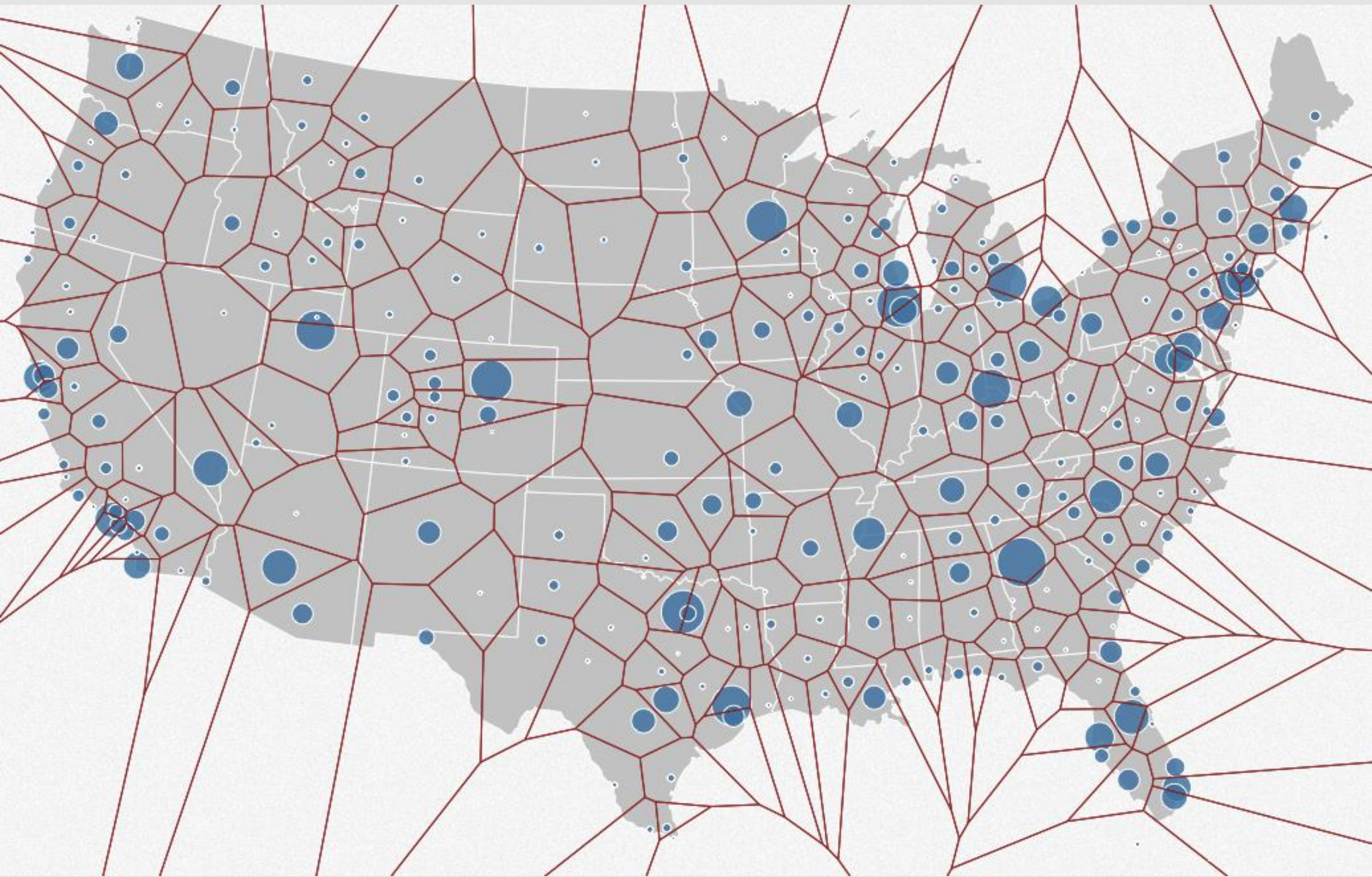
Area Cursors (“Brushes”)

Selection methods

How would you implement this technique? Describe in detail what you

W

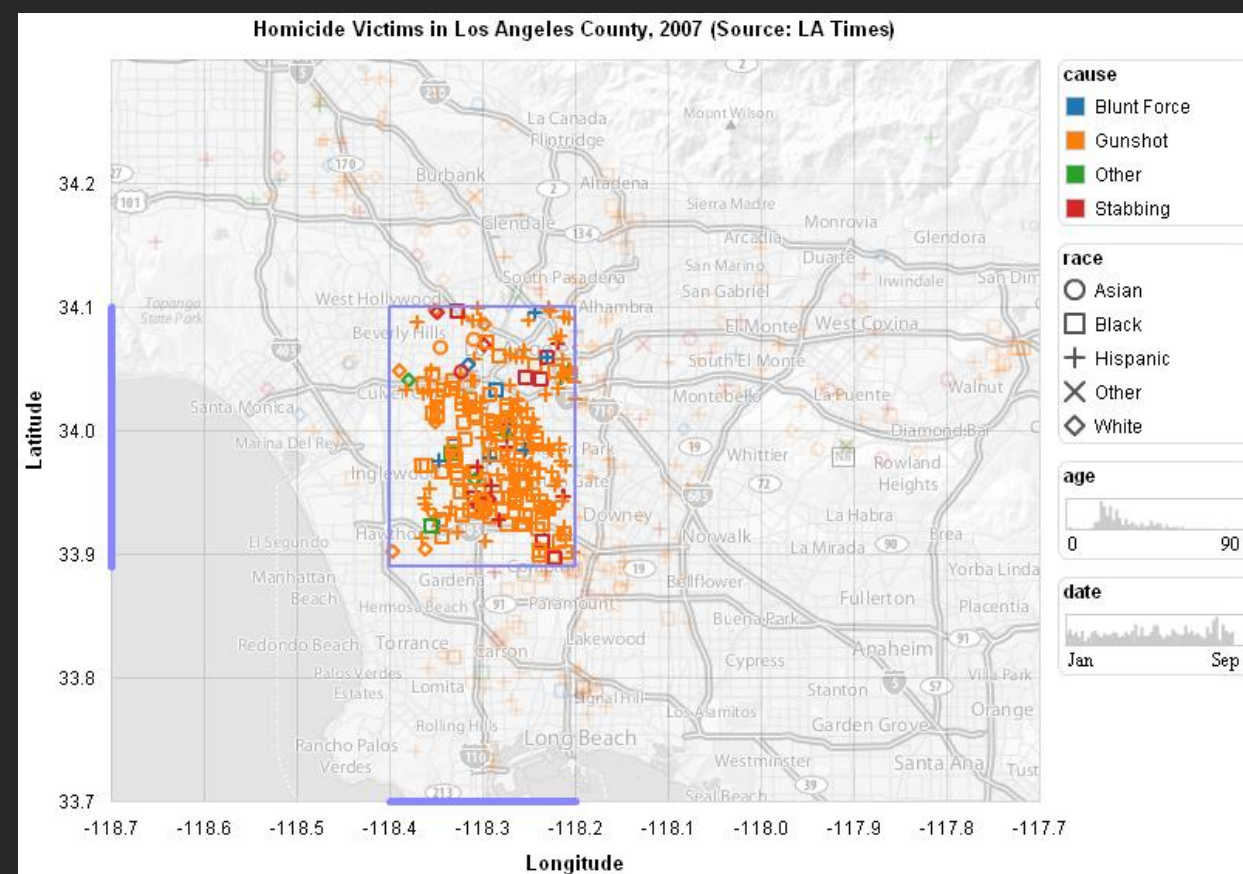




Generalized selection

How is a selection represented?

Model selections as declarative queries



$(-118.371 \leq lon \text{ AND } lon \leq -118.164) \text{ AND } (33.915 \leq lat \text{ AND } lat \leq 34.089)$

Visual Queries

Model selections as declarative queries

Retarget selection across visual encodings

Applicable to dynamic, time-varying data

Perform operations (expansion) on query structure

“Select items like this one.”

Generalized Selection

Point to an example and define an abstraction based on one or more properties
[Clark, Brennan]



“Blue like this”

“The same shape as that”

Abstraction may occur over multiple levels

This is not a sentence.

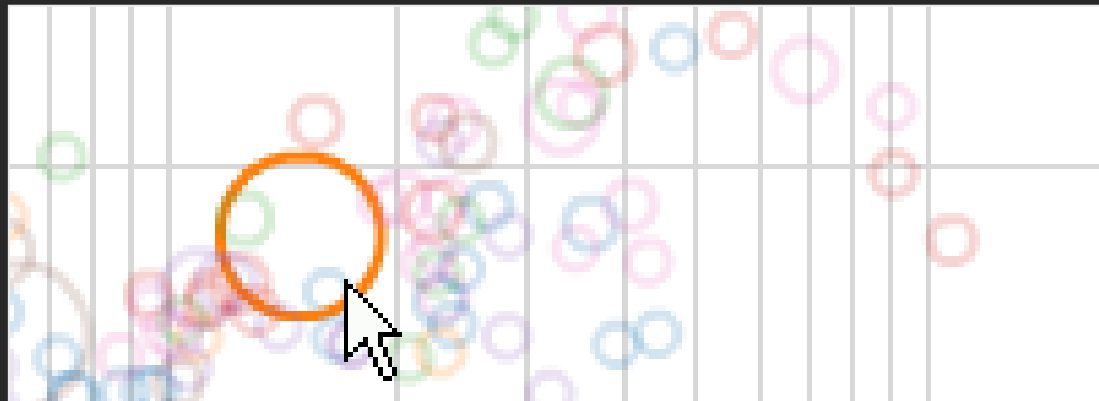


Generalized Selection

Provide *generalization mechanisms* that enable users to *expand a selection query* along *chosen dimensions* of interest

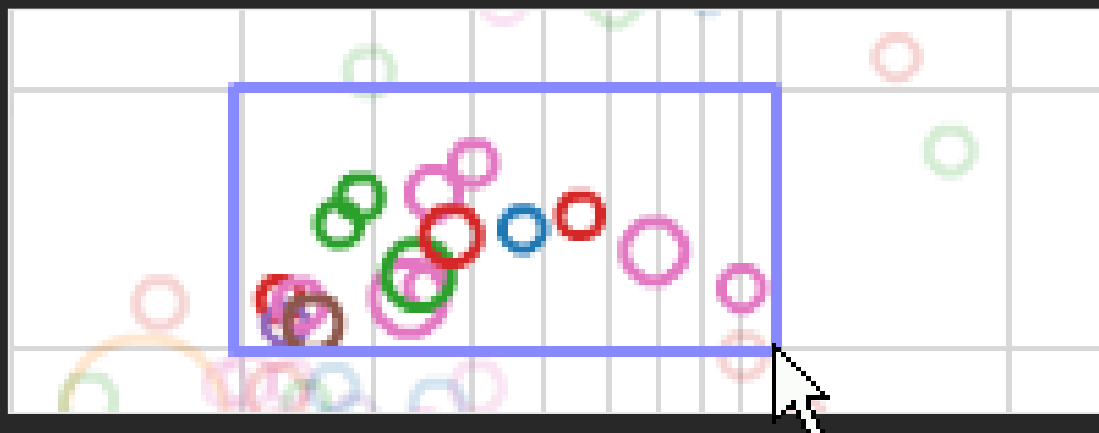
Expand selections via *query relaxation*

Examples



Click: Select Items

`(id = 'China')`



Drag: Select Range

`(2000 < gni AND gni < 10000) AND (.1 < internet AND internet < .2)`



Legend: Select Attributes

`(region = 'The Americas')`

Query Relaxation

Generalize an input query to create an expanded selection, according to:

1. A semantic structure describing the data
2. A traversal policy for that structure

Generalized selection

Generalized Selection via Interactive Query Relaxation

Jeffrey Heer | Maneesh Agrawala | Wesley Willett
University of California, Berkeley

Dynamic Queries

Query and results

SELECT * FROM Houses

WHERE price < 1,000,000 AND bedrooms > 2

ORDER BY price

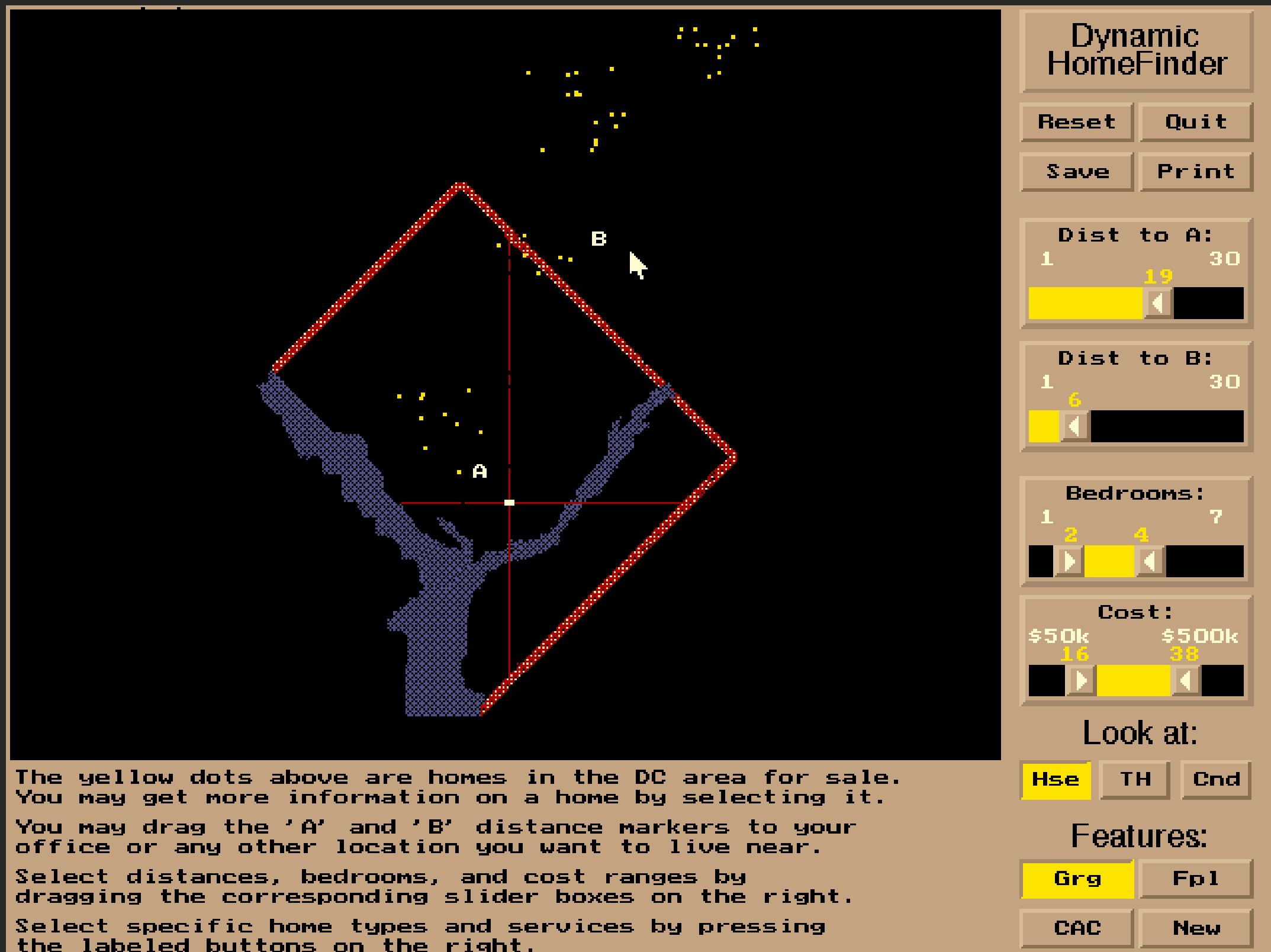
Dynamic Browser : DC Home Finder

IdNumber	Dwelling	Address	City
2	House	5256 S. Capitol St.	Beltsville, MD
4	House	5536 S. Lincoln St.	Beltsville, MD
5	House	5165 Jones Street	Beltsville, MD
8	House	5007 Jones Street	Beltsville, MD
9	House	4872 Jones Street	Beltsville, MD
17	House	5408 S. Capitol St.	Beltsville, MD
20	House	5496 S. Capitol St.	Beltsville, MD
85	Condo	5459 S. Lincoln St.	Laurel, MD
86	Condo	5051 S. Lincoln St.	Laurel, MD
88	Condo	5159 Hamilton Street	Laurel, MD
92	Condo	5132 Hamilton Street	Laurel, MD
93	Condo	5221 S. Lincoln St.	Laurel, MD
94	Condo	5043 S. Lincoln St.	Laurel, MD
95	Condo	4970 Jones Street	Laurel, MD
97	Condo	4677 Jones Street	Laurel, MD
98	Condo	4896 S. Capitol St.	Laurel, MD
99	Condo	5048 S. Capitol St.	Laurel, MD
100	Condo	4597 31st Street	Laurel, MD
101	Condo	5306 S. Lincoln St.	Laurel, MD
103	Condo	5562 Glass Road	Laurel, MD
105	Condo	5546 Hamilton Street	Laurel, MD
152	House	7670 31st Street	Upper Marlboro, MD

Issues

1. For programmers
2. Rigid syntax
3. Only shows exact matches
4. Too few or too many hits
5. No hint on how to reformulate the query
6. Slow question-answer loop
7. Results returned as table

HomeFinder



The yellow dots above are homes in the DC area for sale. You may get more information on a home by selecting it. You may drag the 'A' and 'B' distance markers to your office or any other location you want to live near. Select distances, bedrooms, and cost ranges by dragging the corresponding slider boxes on the right. Select specific home types and services by pressing the labeled buttons on the right.

Dynamic HomeFinder

Reset Quit

Save Print

Dist to A:
1 30
19

Dist to B:
1 30
6

Bedrooms:
1 7
2 4

Cost:
\$50k \$500k
16 38

Look at:
Hse TH Cnd

Features:
Grp Fp1
CAC New

[Ahlberg and Schneiderman 92]

Direct manipulation

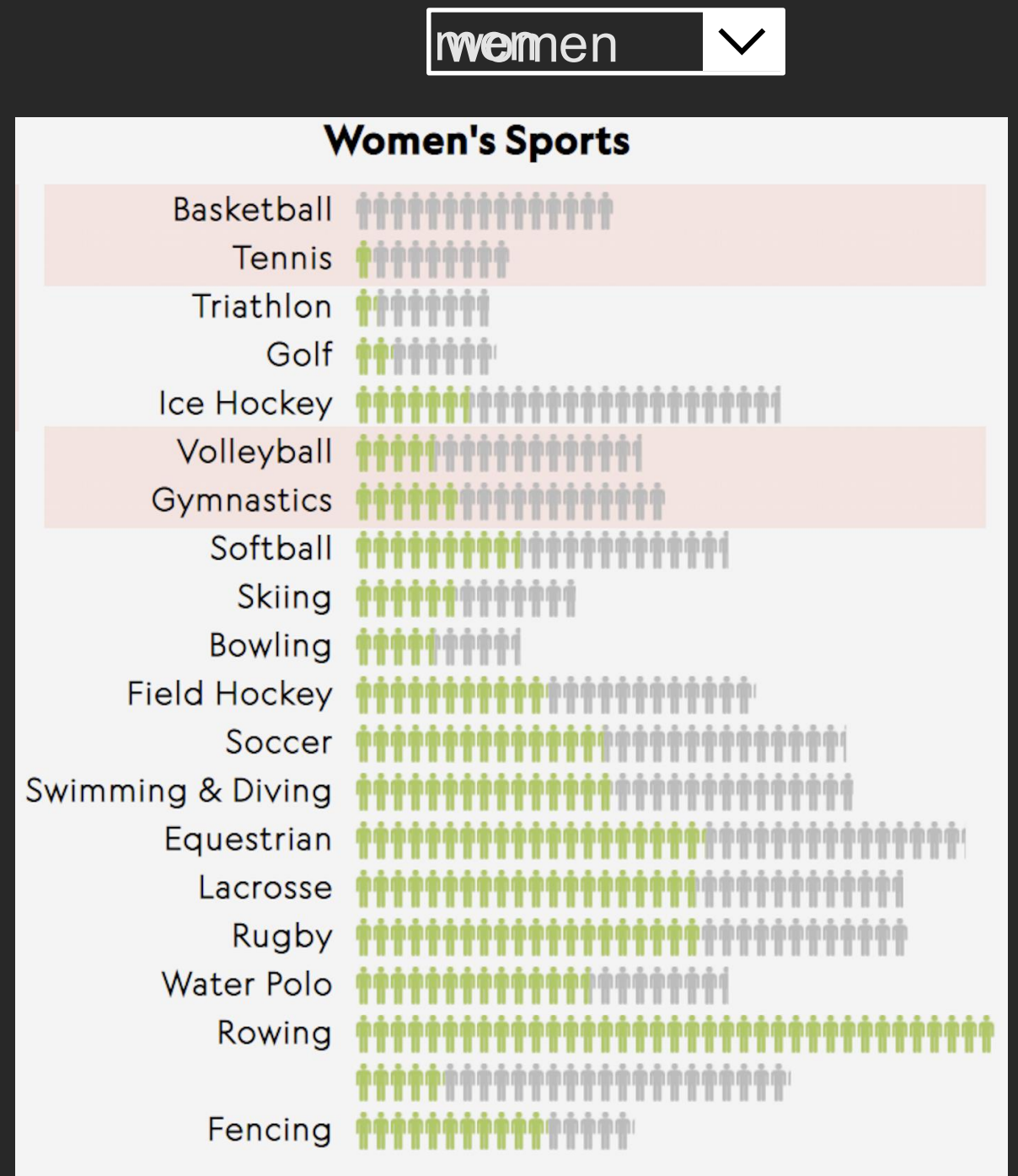
1. Visual representation of objects and actions
2. Rapid, incremental and reversible actions
3. Selection by pointing (not typing)
4. Immediate and continuous display of results

Dynamic queries & comparison

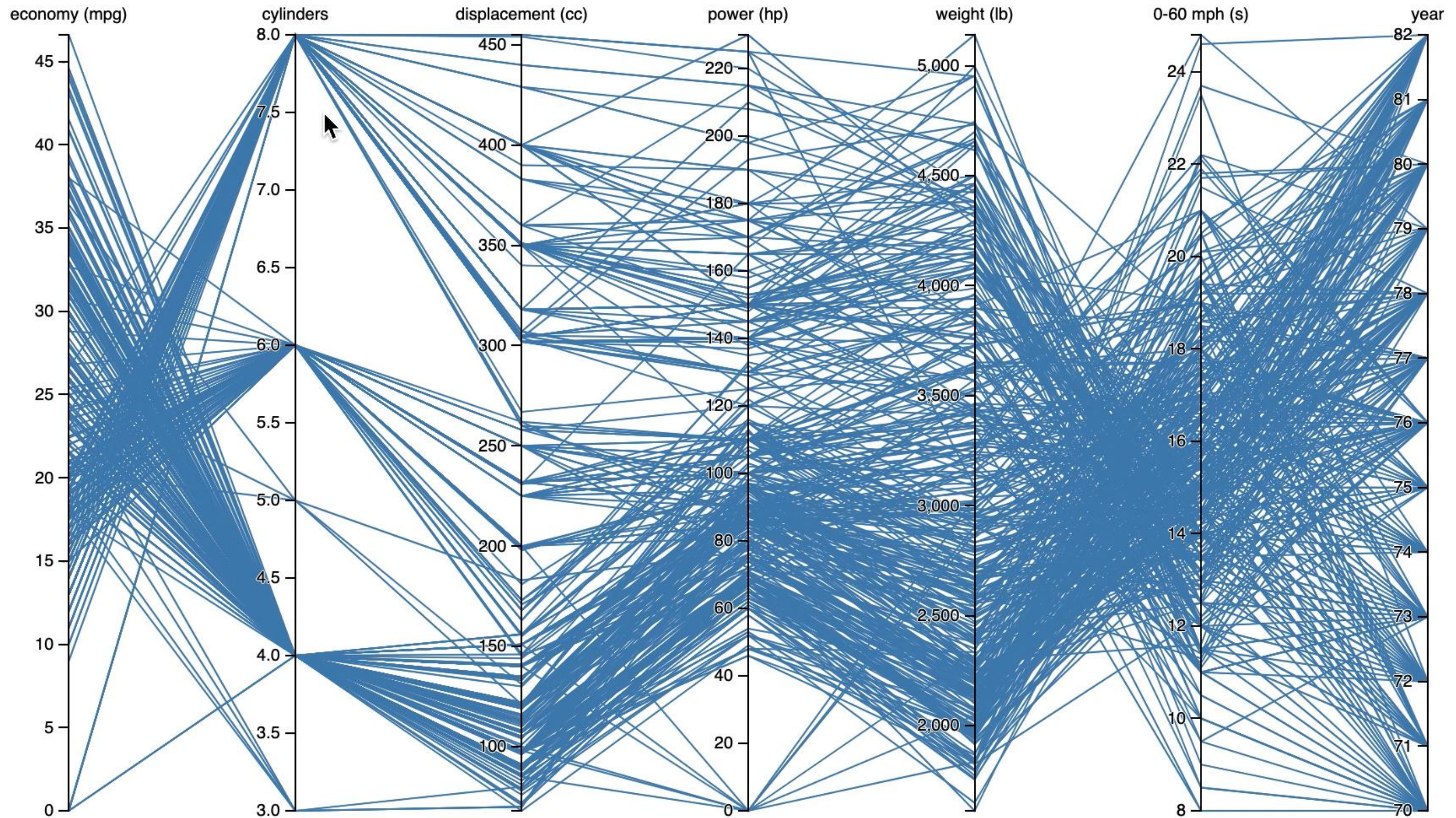
Comparing data is one of the most important functions of a visualization.

What if I want to compare the results of multiple dynamic queries?

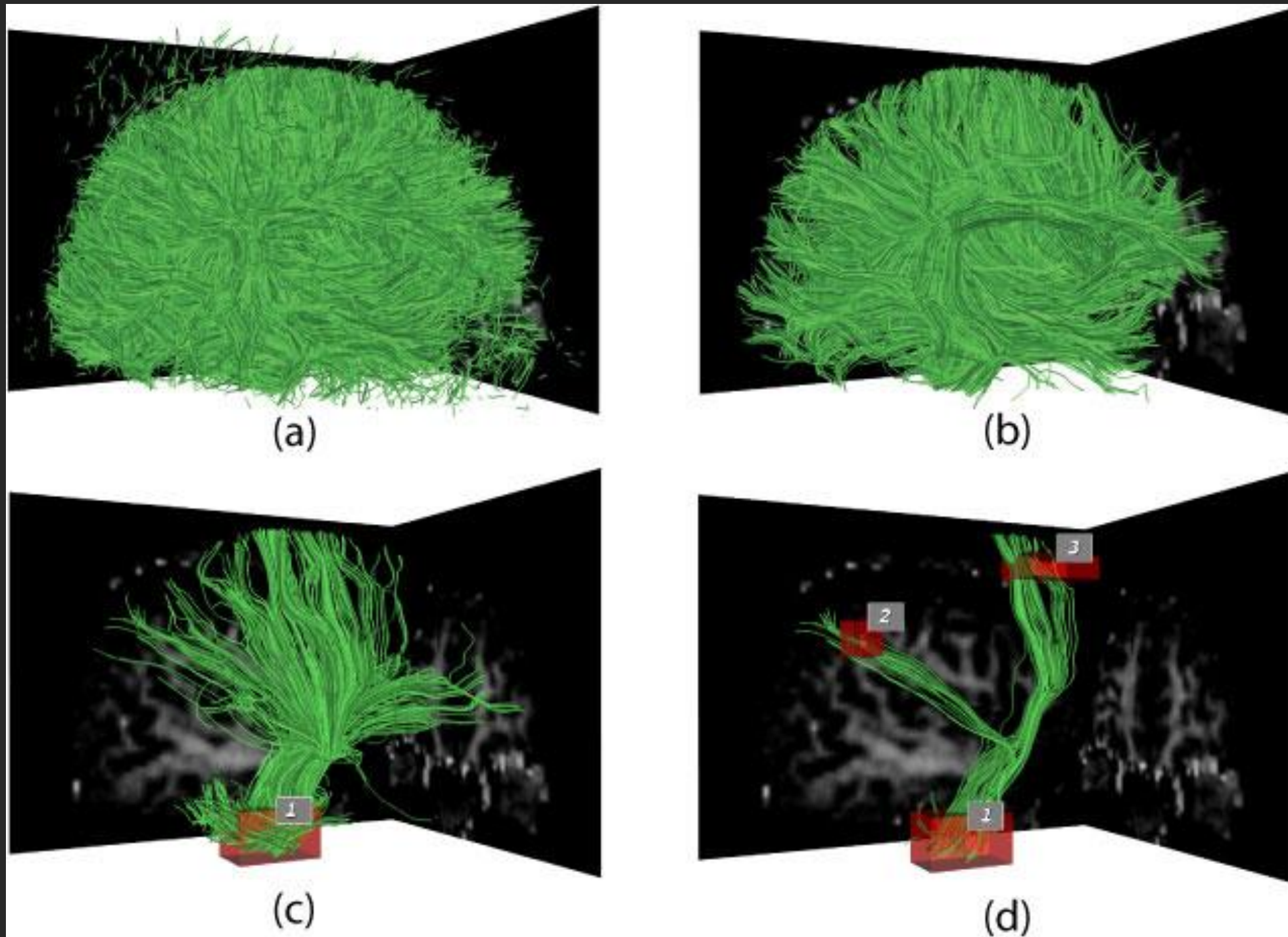
“Eyes beat memory every time.”



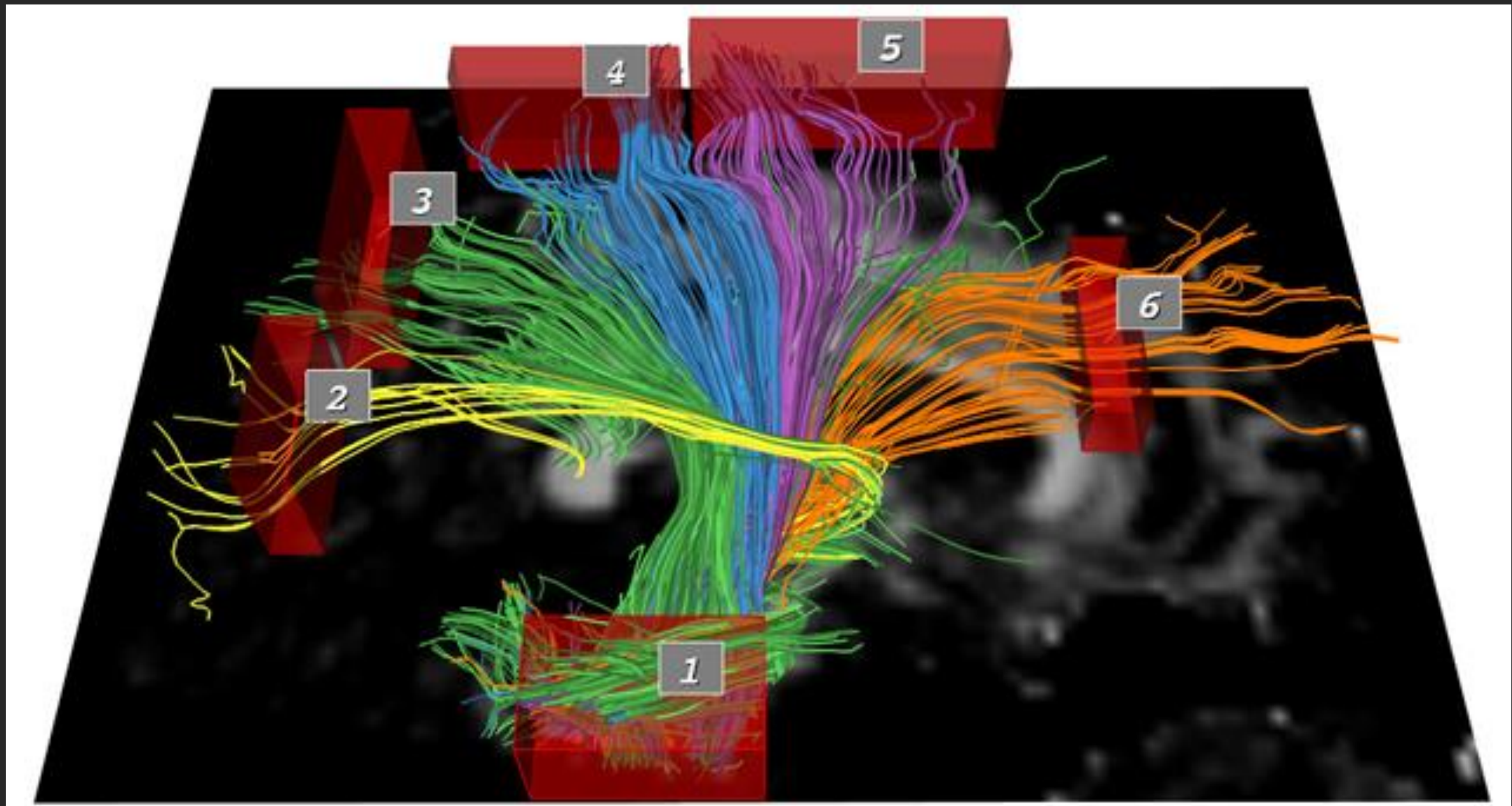
Interactive parallel coordinates



3D dynamic queries [Akers et al. 04]



3D dynamic queries [Akers et al. 04]



Summary

Most visualizations are interactive

- ❑ Even passive media elicit interactions

Good visualizations are task dependant

- ❑ Choose the right use of space
- ❑ Pick the right interaction technique

Human factors are important

- ❑ Leverage human strengths
- ❑ Assist to get past human limitations

Animation!

Goals for animation use

Help visualization convey information

How does *animation* help convey information?

- Explain a process
- Smooth a transition between states

Social Mobility [New York Times 2018]

Follow the lives of white boys and black boys from poor households, using their household incomes as adults, and including kids of all mothers, including immigrants :

Follow the lives of these **12,579** Americans and see where they end up as adults:



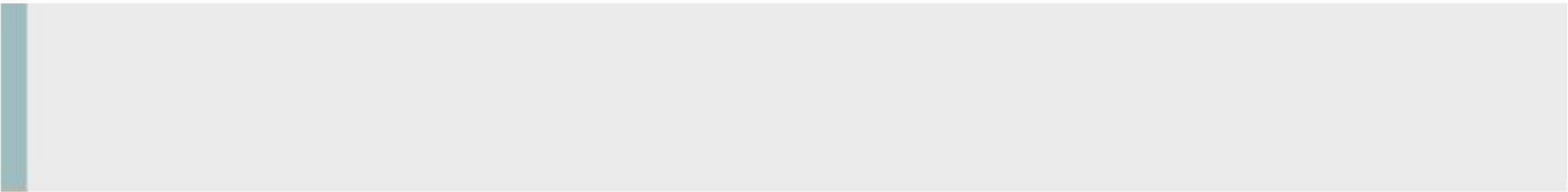
Covid-19

[Washington Post 2020]

Count

Recovered	0
Healthy	199
Sick	1

Change over time



Free-for-all

Attempted quarantine

Moderate distancing

Extensive distancing



NameVoyager [Wattenberg 04]

NameVoyager: Explore baby names and name trends letter by letter

Looking for the perfect baby name? **Sign up for free** to receive access to our expert tools!

Baby Name >



Both



Boys



Girls

Current rank:

boys

1000

500

100

25

1

girls

1000

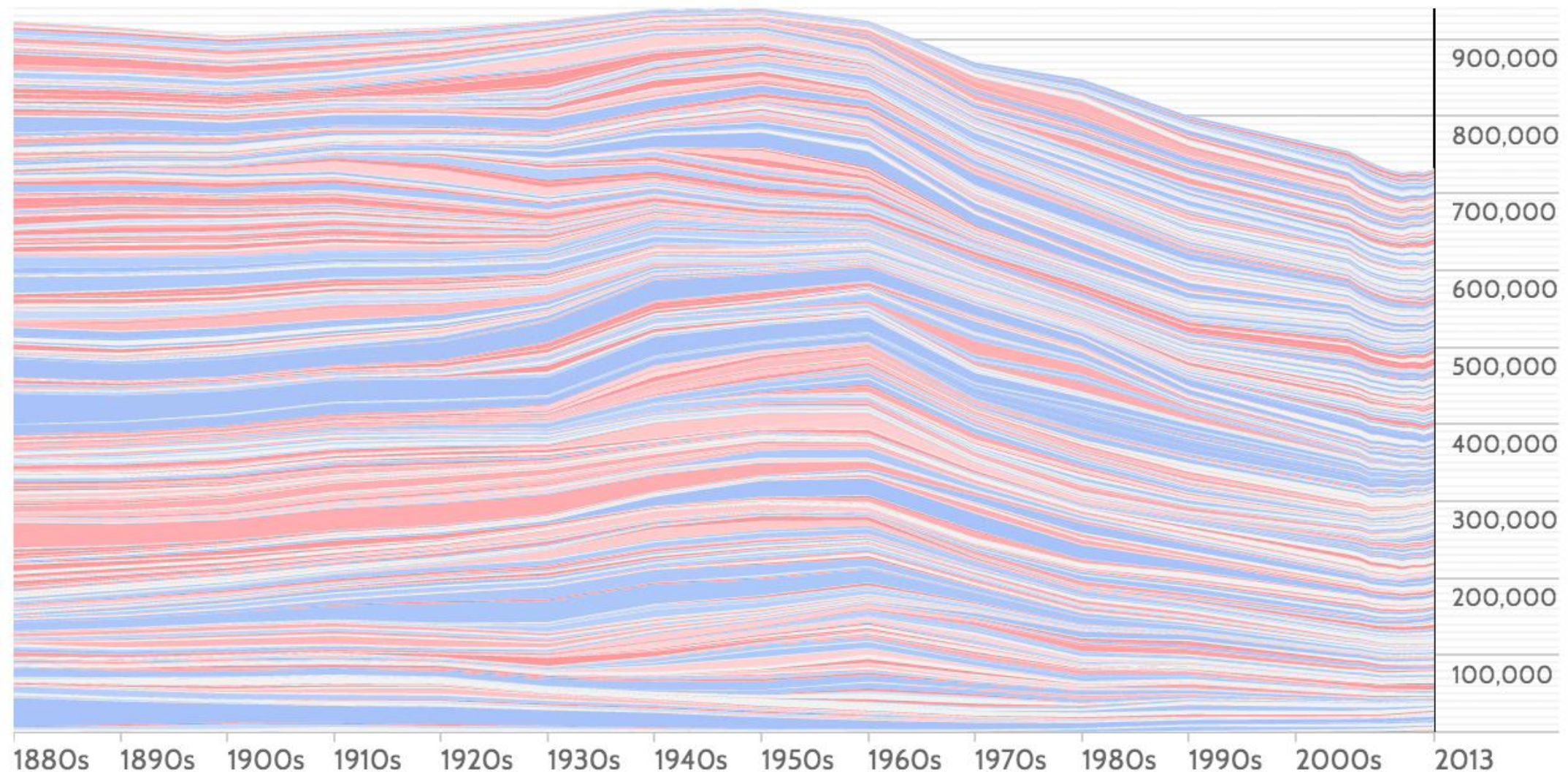
500

100

25

1

per million births



Click a name graph to view that name. Double-click to read more about it.

[enlarge](#)

<http://www.babynamewizard.com/namevoyager/Inv0105.html>

Motions show transitions

Can see change from one state to next

- States are spatial layouts
- Changes are simple transitions (translation, rotation, scale)



start

end

Shows transition better, but

- Still may be too fast, or too slow
- Too many objects may move at once

Problems [Tversky 02]

Difficulties in understanding animation

- ❑ Difficult to estimate paths and trajectories
- ❑ Motion is fleeting and transient
- ❑ Cannot simultaneously attend to multiple motions
- ❑ Trying to parse motion into events, actions and behaviors
- ❑ Misunderstanding and wrongly inferring causality
- ❑ Anthropomorphizing physical motion may cause confusion or lead to incorrect conclusions

Principles for conveying information

Congruence:

The structure and content of the external representation should correspond to the desired structure and content of the internal representation.

Apprehension:

The structure and content of the external representation should be readily and accurately perceived and comprehended.

Principles for Animation

Congruence

- Maintain valid data graphics during transitions
- Use consistent syntactic/semantic mappings
- Respect semantic correspondence
- Avoid ambiguity

Apprehension

- Group similar transitions
- Minimize occlusion
- Maximize predictability
- Use simple transitions
- Use staging for complex transitions
- Make transitions as long as needed, but no longer
- Introduce interactivity to help user avoid difficulties

Goals for animation use

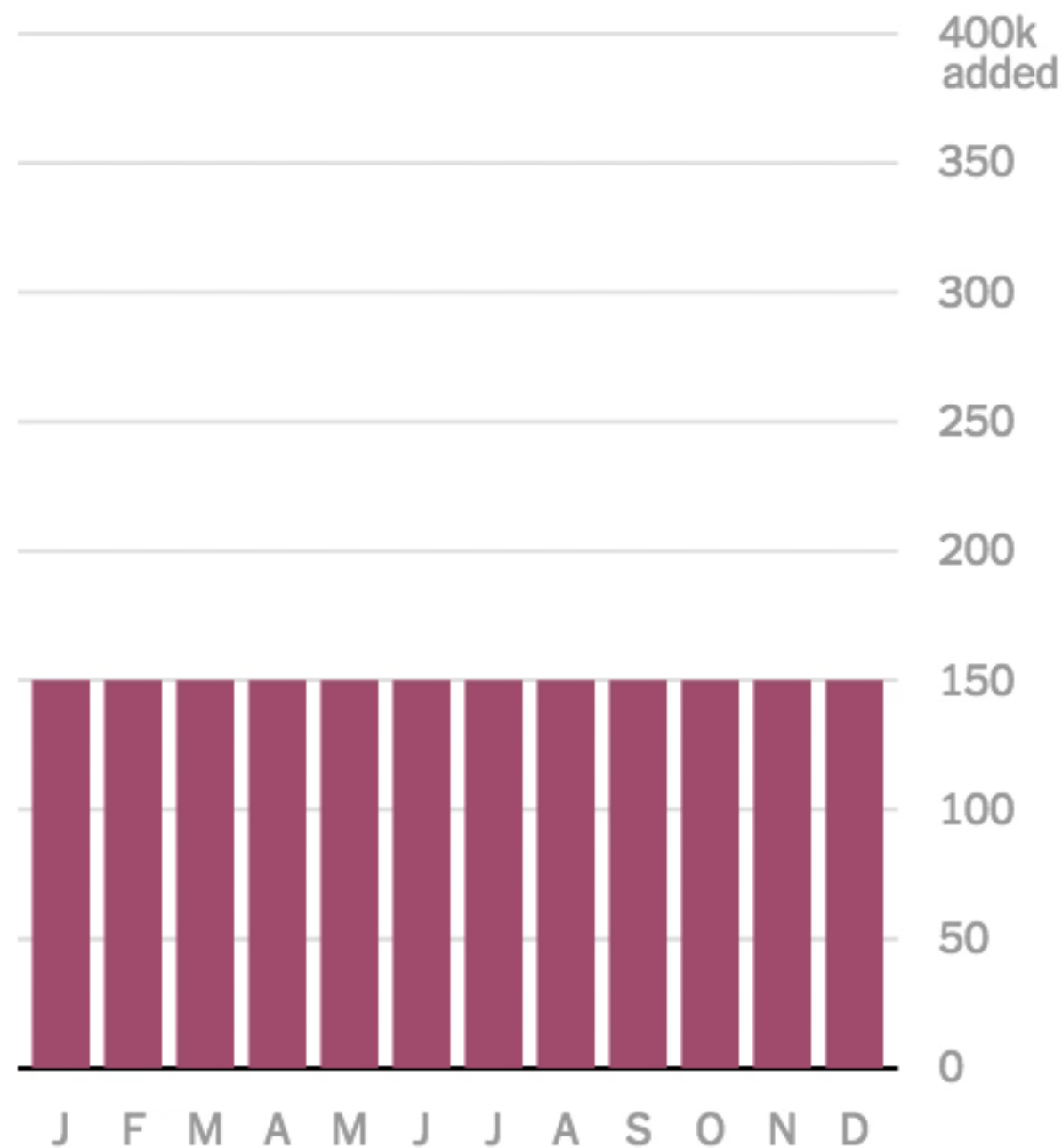
Help visualization convey information

How does *animation* help convey information?

- Explain a process
- Smooth a transition between states
- Present alternative possible realizations (show uncertainty)

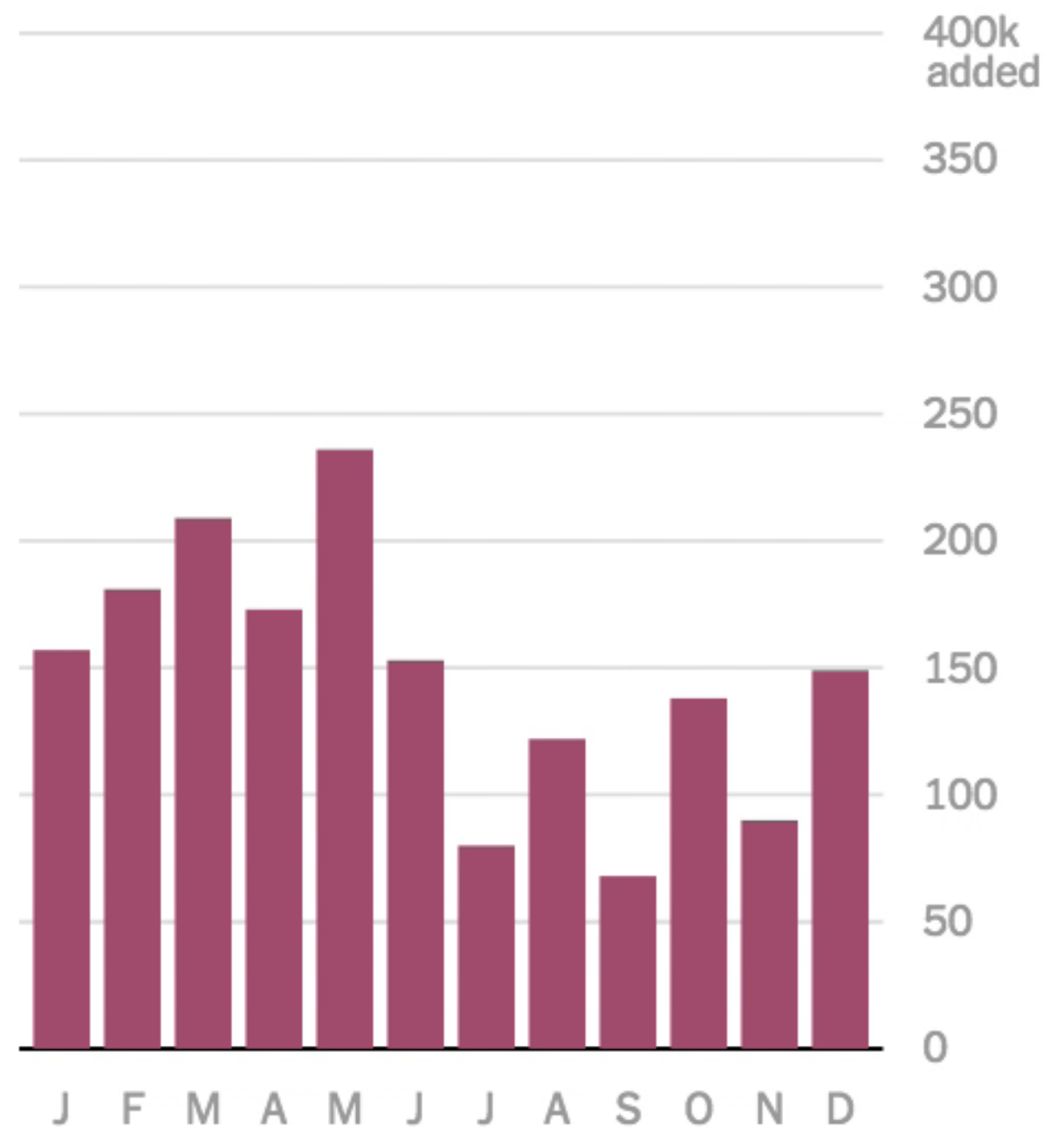
Visualizing uncertainty

If job growth **were actually steady** over the last 12 months...



...the jobs report **could look like this:**

Pause



Understanding Motion

Motion as a visual cue

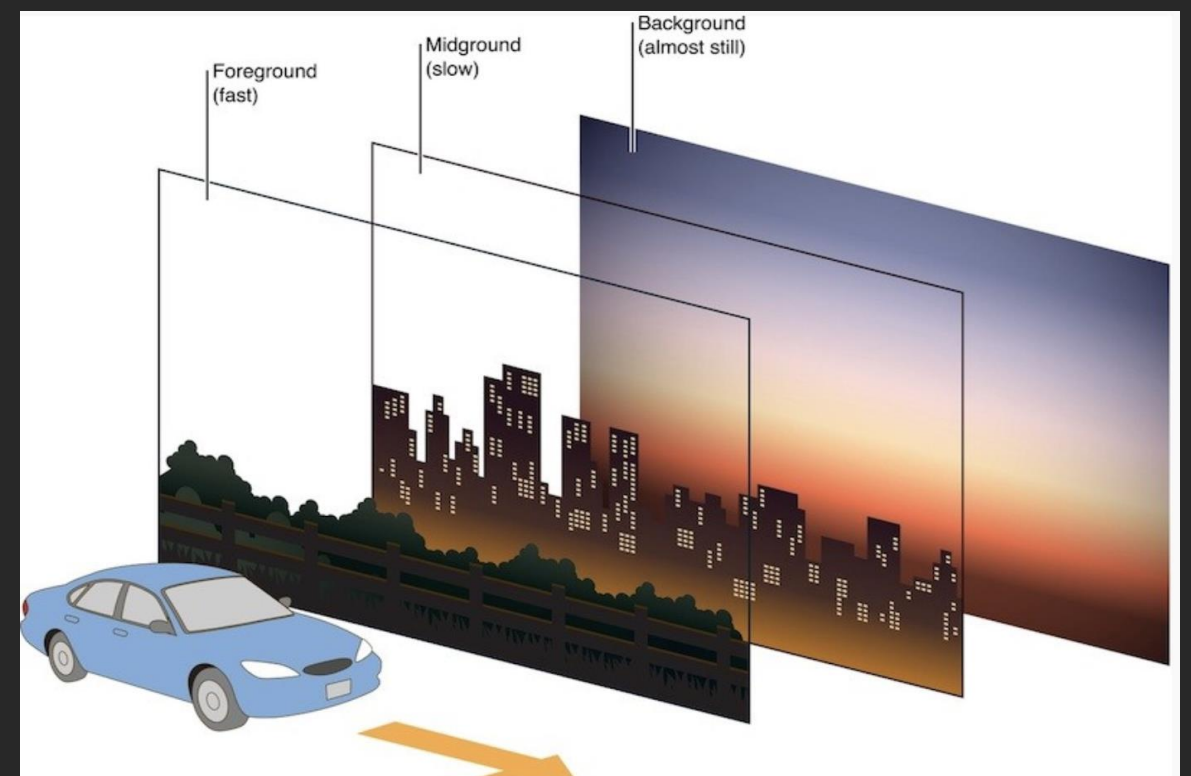
“Pre-attentive”

- ❑ Stronger than color, shape, ...

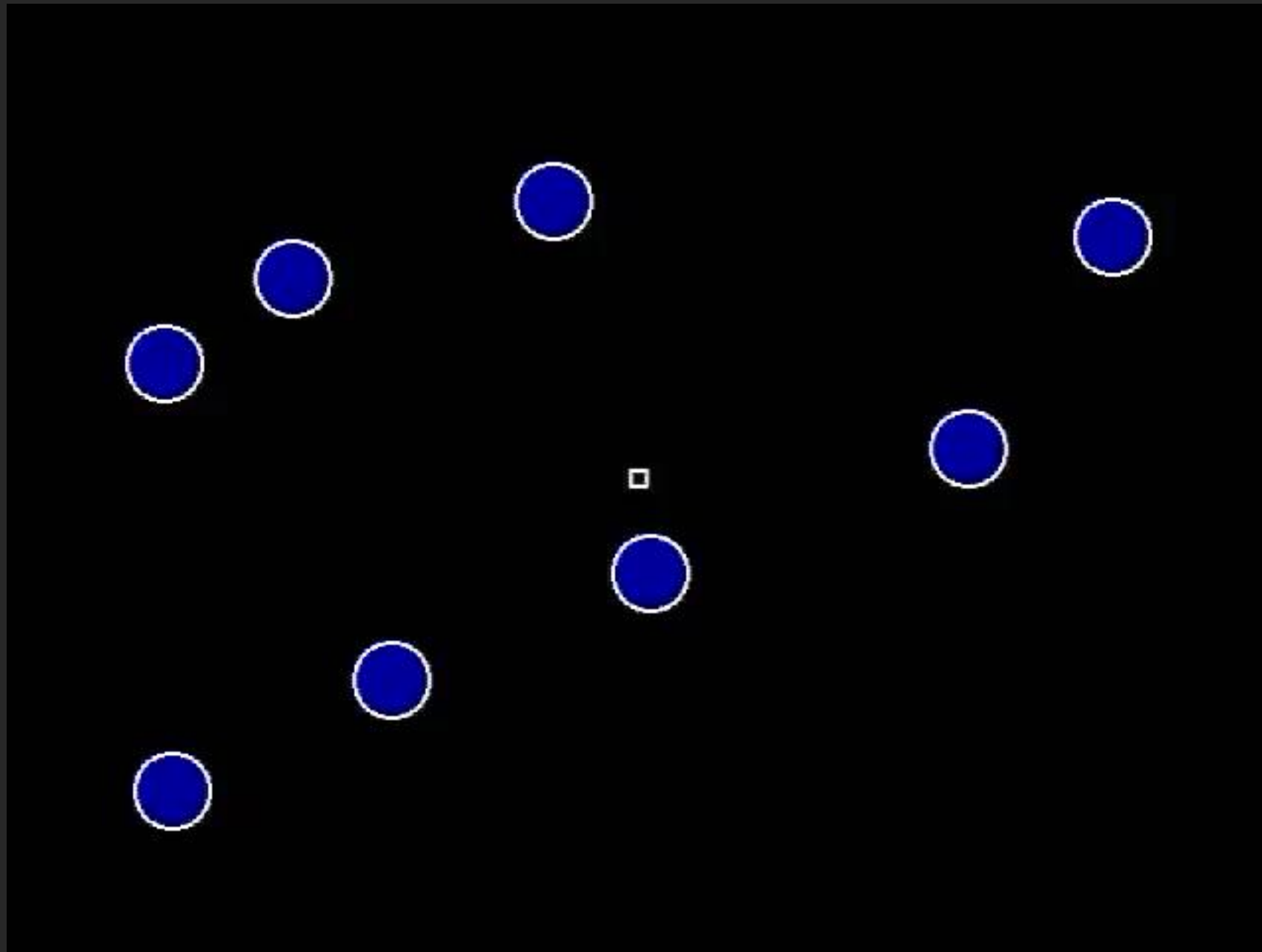
More sensitive to motion at periphery

Triggers an orientation response

Motion parallax provides 3D cue (like stereopsis)

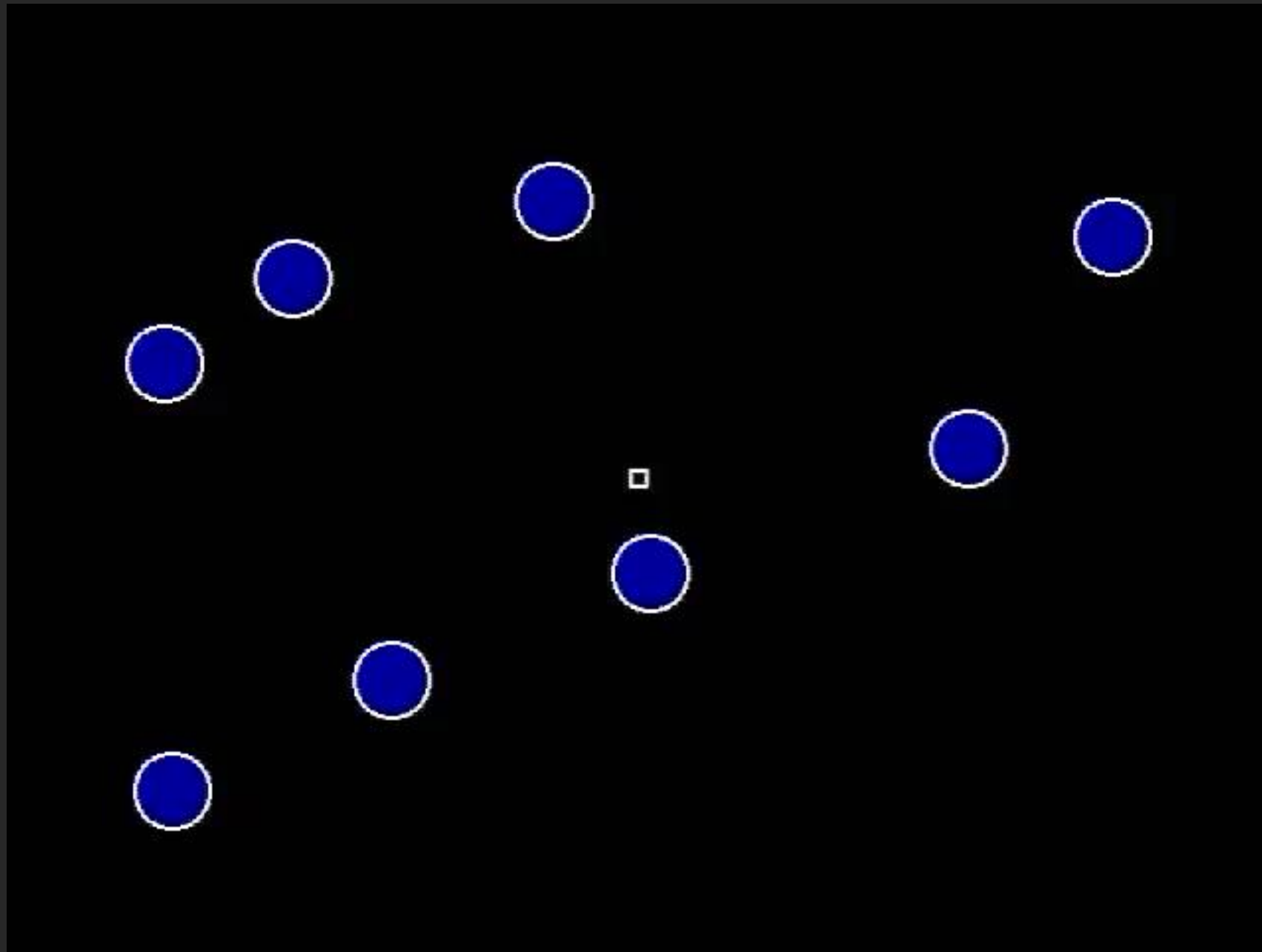


Tracking multiple targets



How many dots can we simultaneously track?

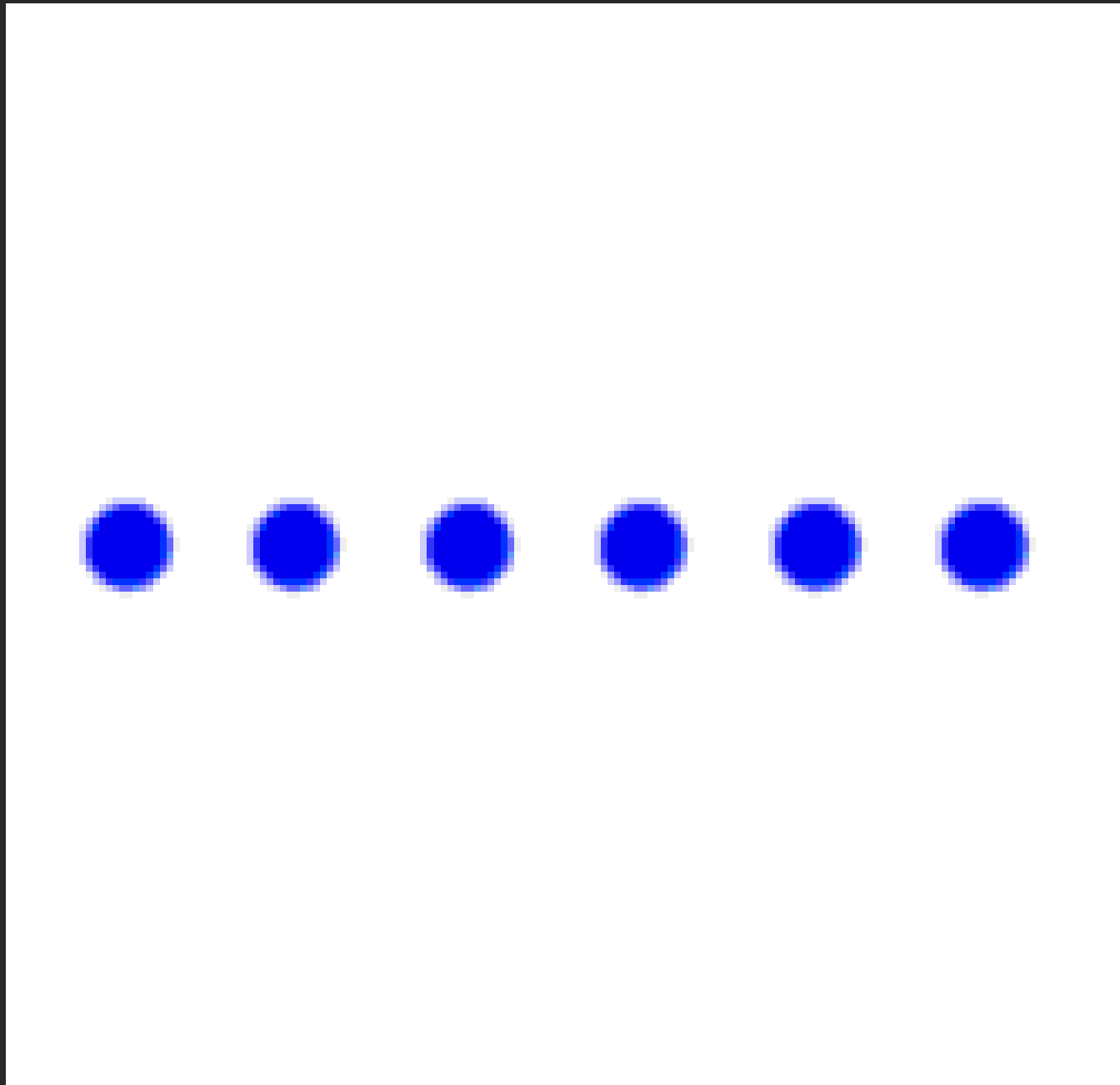
Tracking multiple targets



How many dots can we simultaneously track?

□ 4 to 6 - difficulty increases significantly at 6

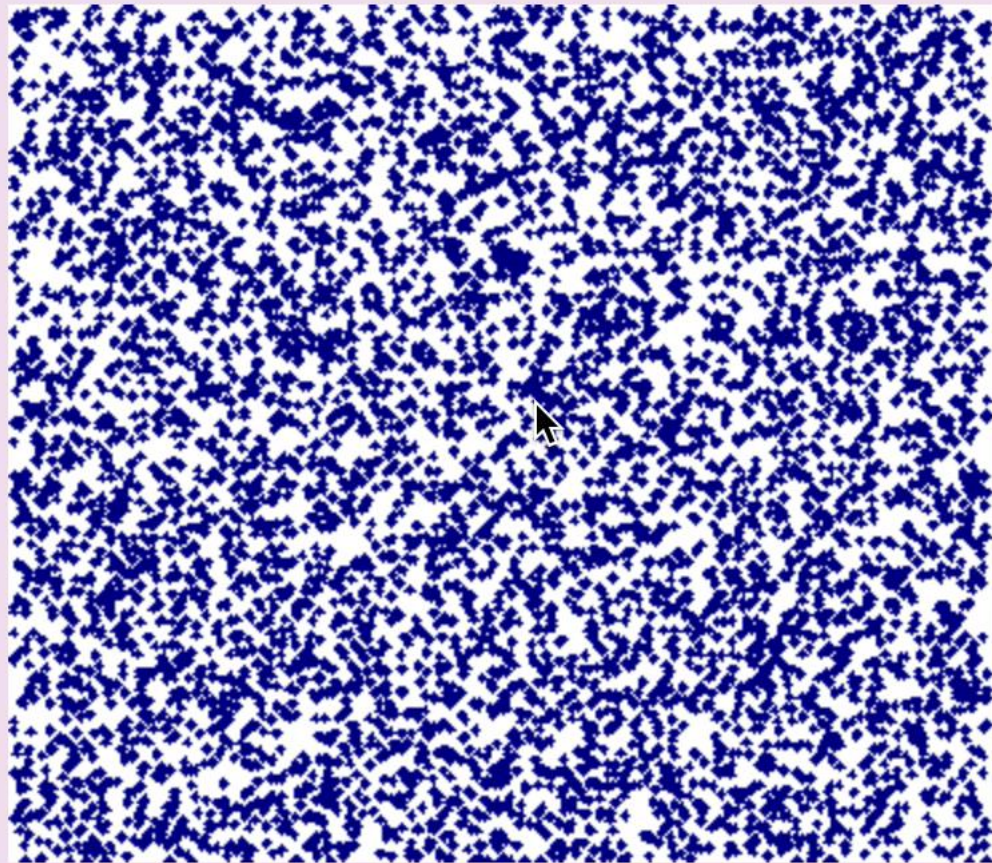
Grouped dots count as 1 object



Dots moving together are grouped

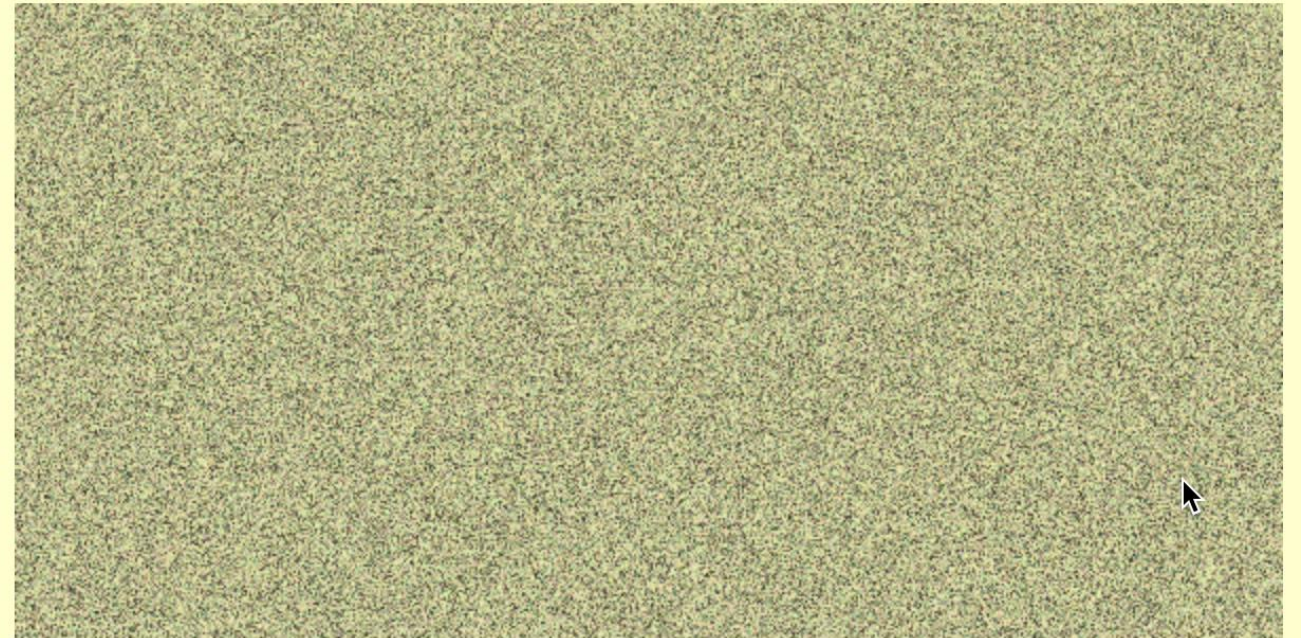
<http://coe.sdsu.edu/eet/articles/visualperc1/start.htm>

Segment by common motion (fate)



Sand Shrimp

These camouflaged creatures are shy and prefer to hide.
They reveal themselves only when they feel a nudge.

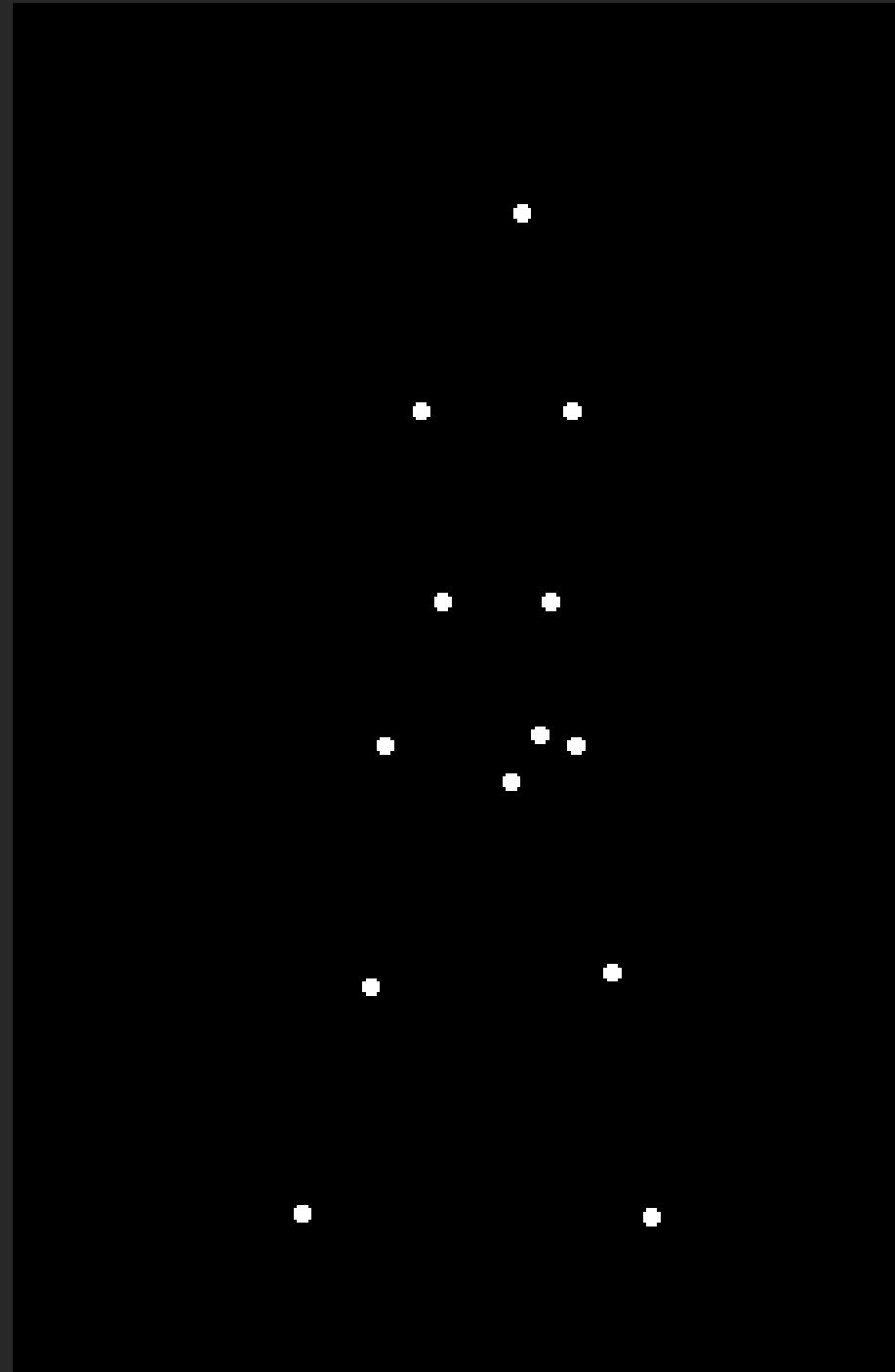


singlecell: July 2001
by Martin Wattenberg, New York
See also: [The Shape of Song](#) - [Apartment](#) - [Map of the Market](#)

<http://dragon.uml.edu/psych/commfate.html>

<http://www.singlecell.org/july/index.html>

Grouping based on biological motion



[Johansson 73]

Constructing narratives

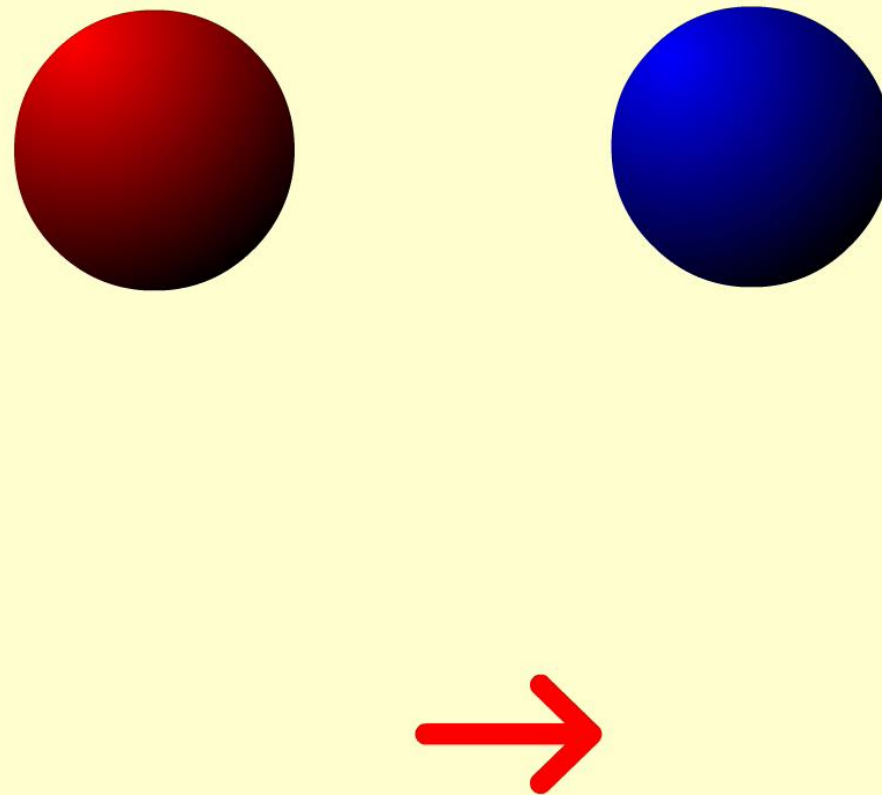
Animation from:
Heider, F. & Simmel, M. (1944).
An experimental study of apparent behavior.
American Journal of Psychology, 57, 243-259.

Courtesy of:
Department of Psychology,
University of Kansas, Lawrence.

http://anthropomorphism.org/img/Heider_Flash.swf

Attribution of causality [Michotte 46]

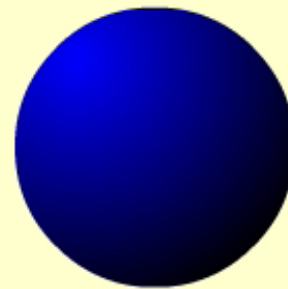
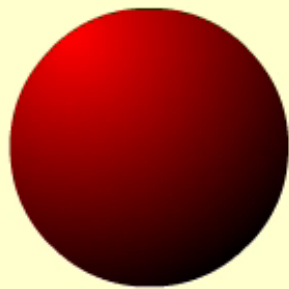
Michotte demonstration 1. What do you see? Most observers report that "the **red ball** hit the **blue ball**." The **blue ball** moved "because the **red ball** hit it." Thus, the **red ball** is perceived to "cause" the **blue ball** to move, even though the balls are nothing more than color disks on your screen that move according to a programme.



http://cogweb.ucla.edu/Discourse/Narrative/Heider_45.html

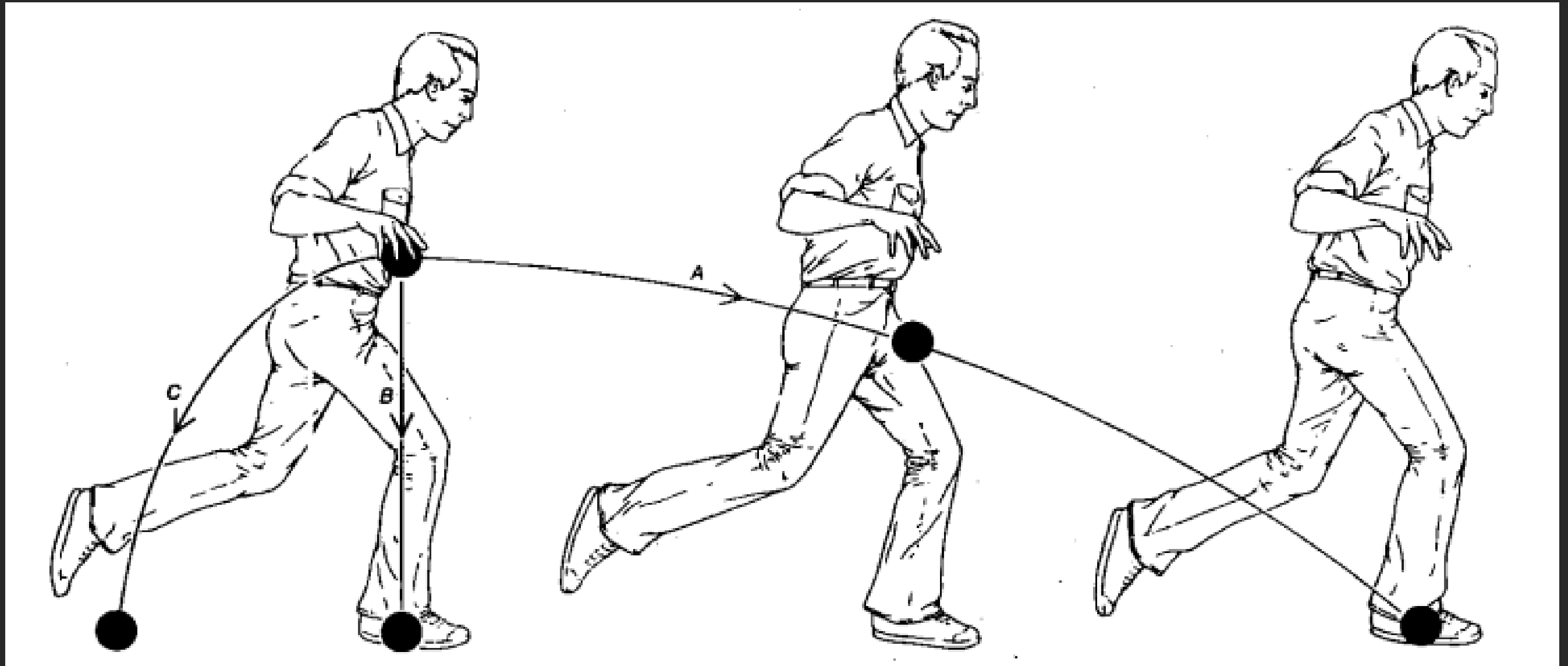
Attribution of causality [Michotte 46]

Michotte demonstration 2. What do you see? Most observers report that "The **red ball** moved up to **blue ball**, and stopped. Then the **blue ball** moved." No cause and effect is implied; the one-second delay between the two movements has reduced the perception of causality.



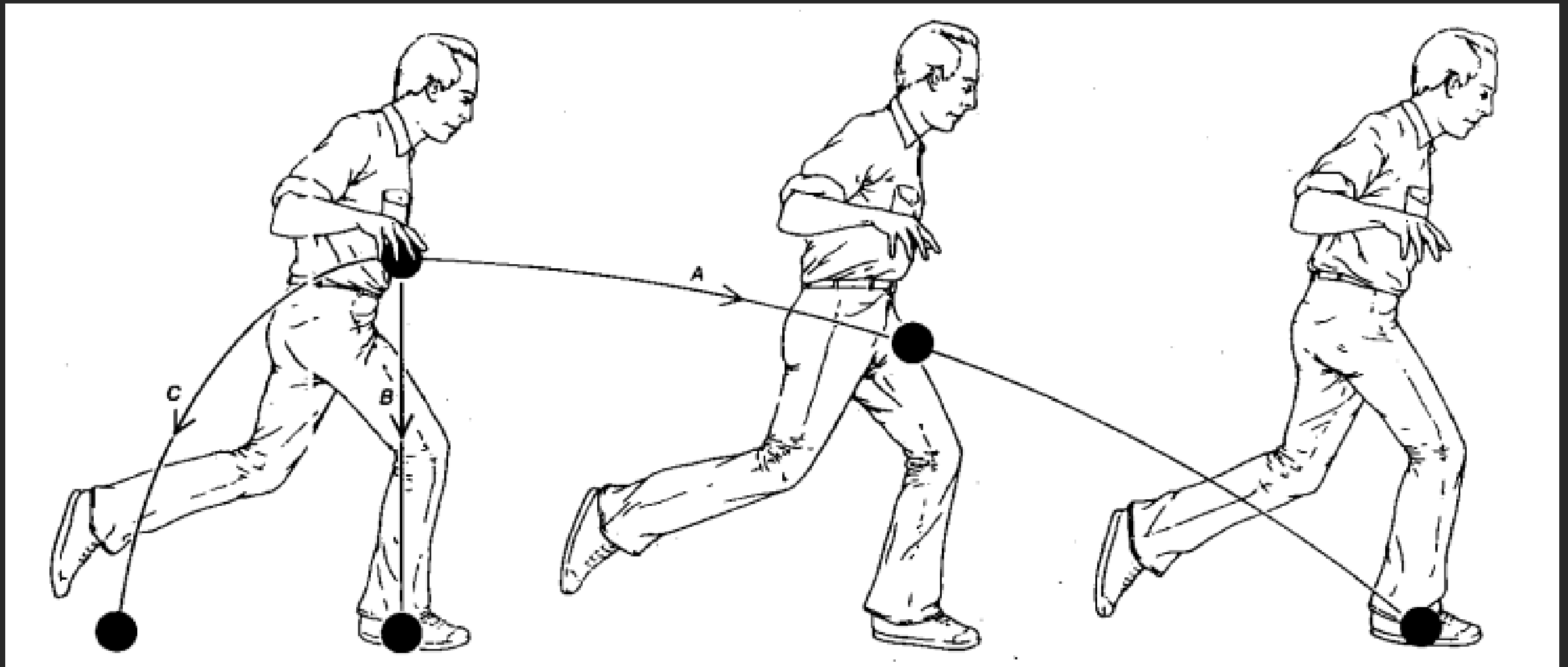
http://cogweb.ucla.edu/Discourse/Narrative/Heider_45.html

Intuitive physics [McCloskey 83]



Running man drops ball. What is the trajectory of the ball?

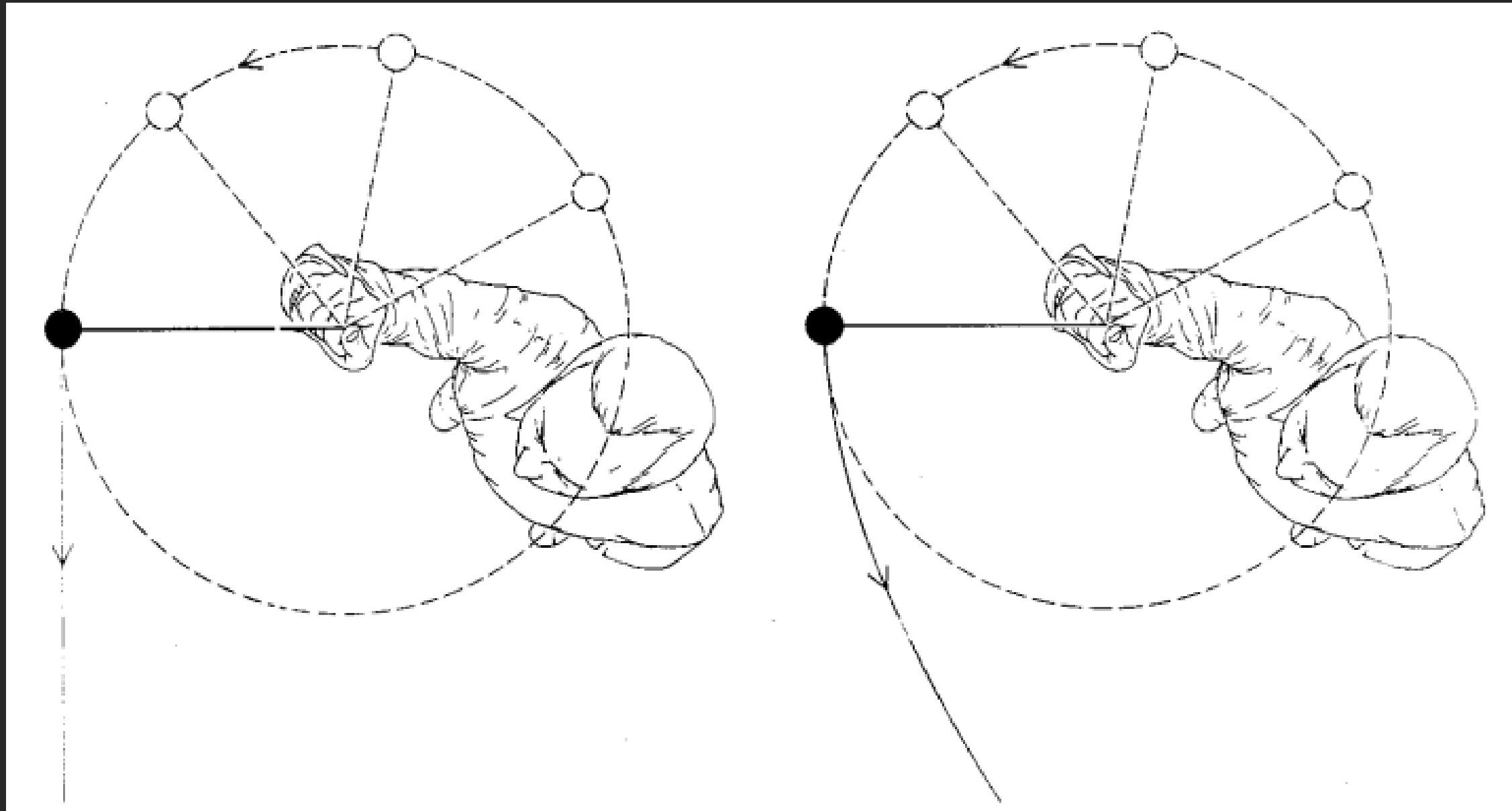
Intuitive physics [McCloskey 83]



Running man drops ball. What is the trajectory of the ball?

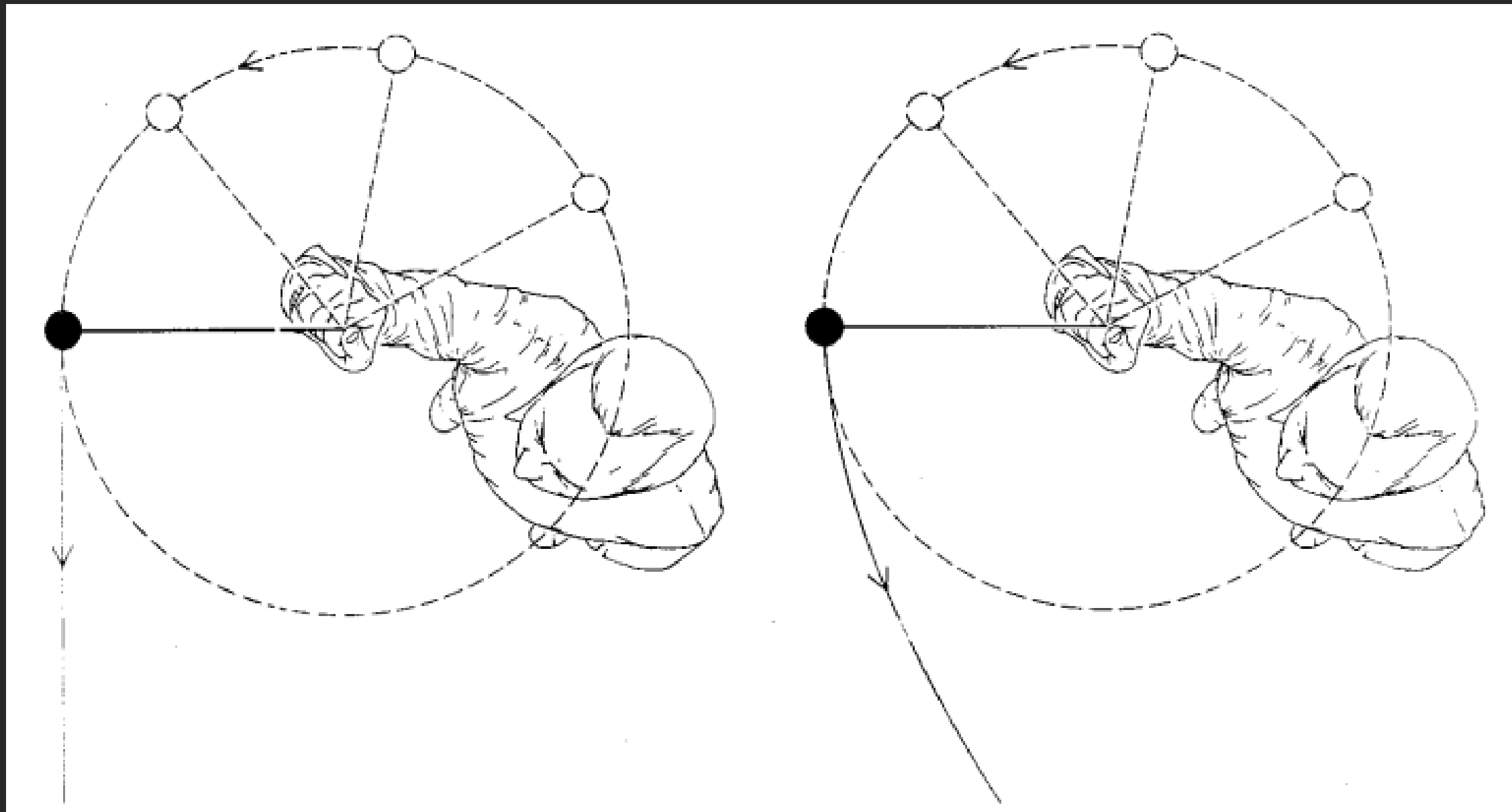
College students: Straight down (49%) , Bkwd (6%), Fwd (45%)

Intuitive physics [McCloskey 83]



Man is swinging ball on end of string. String is cut.
Draw trajectory of the ball.

Intuitive physics [McCloskey 83]



**Man is swinging ball on end of string. String is cut.
Draw trajectory of the ball.**

51% Draw correct path
30% Draw curved path
19% Draw other incorrect paths

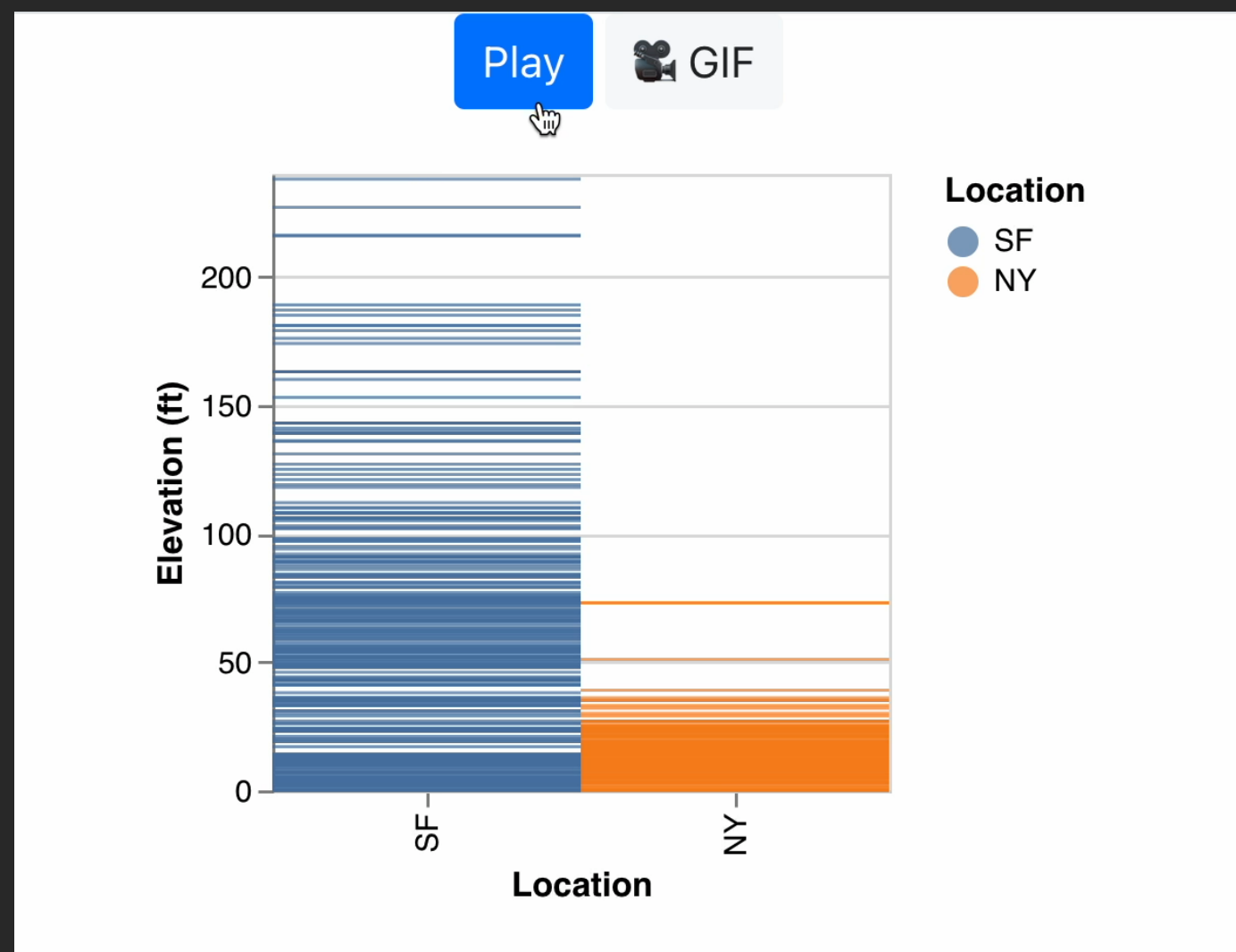
Animated transitions in vis

Main choices:

Start and end states

Keyframes/staging

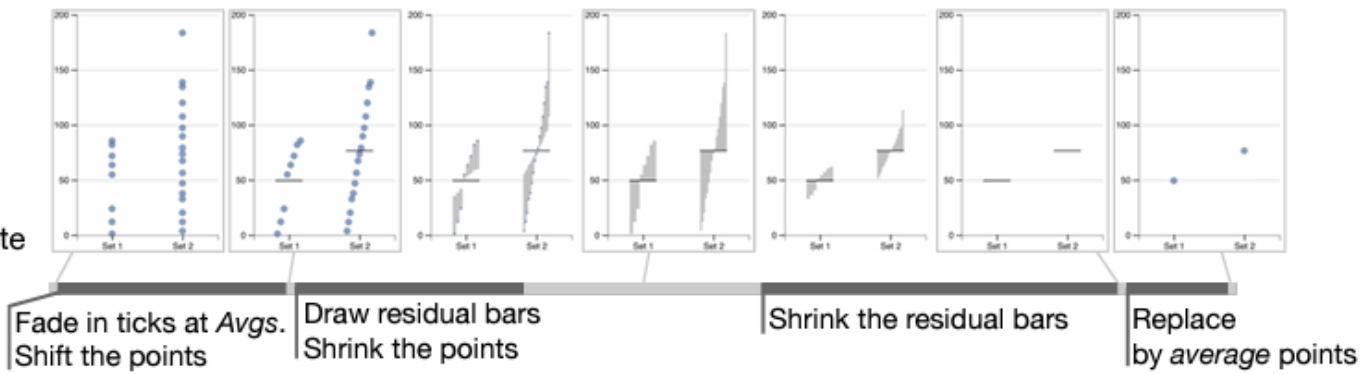
Easing functions



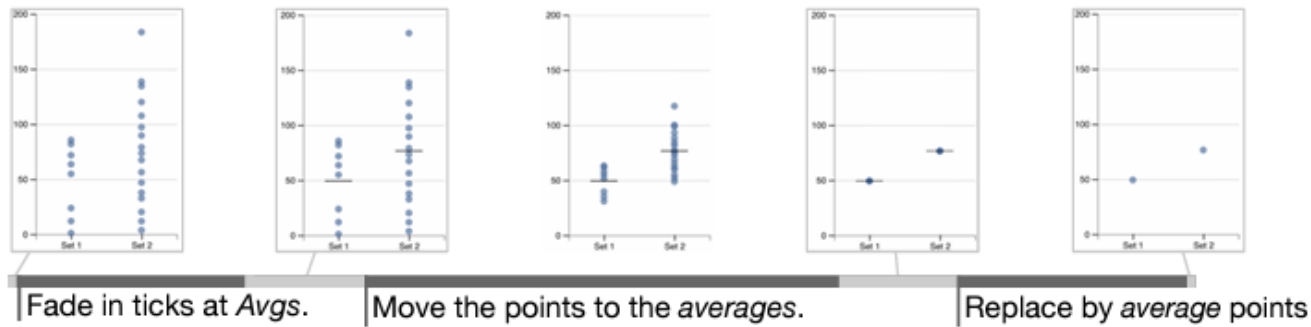
Conveying aggregations

Average

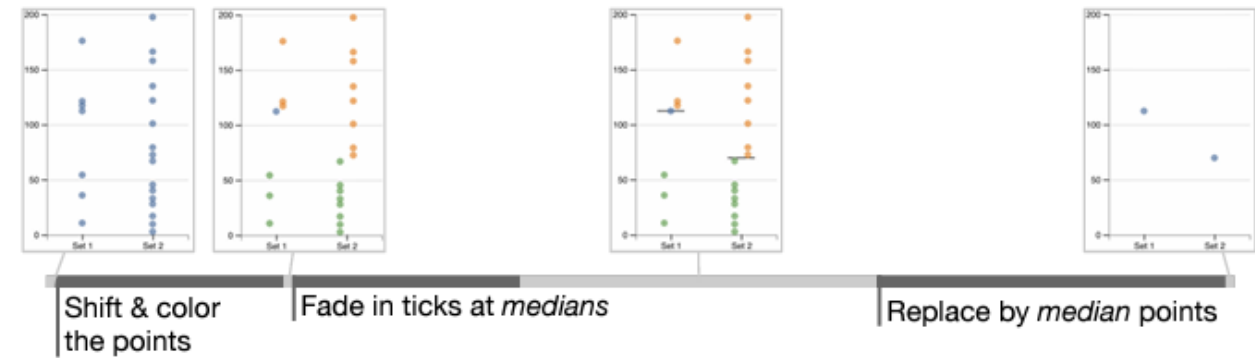
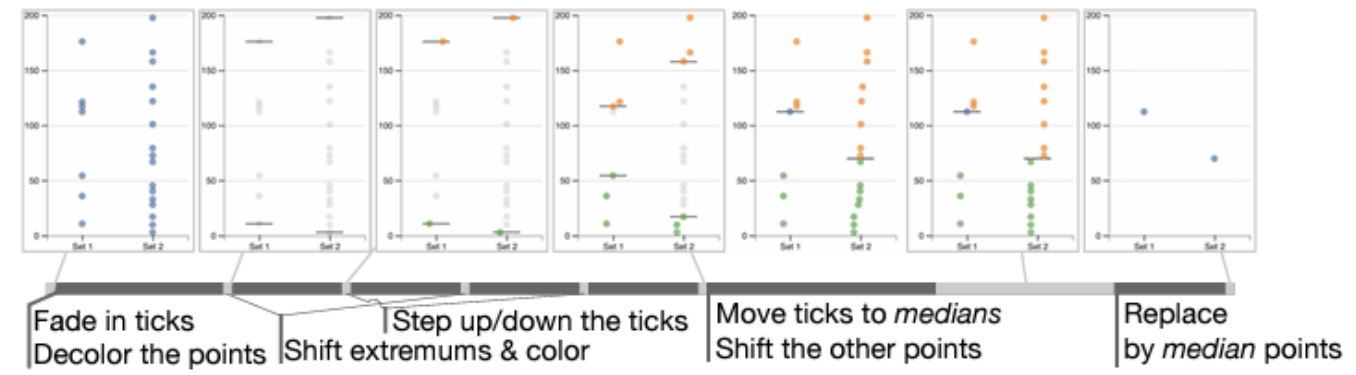
Staged
Elaborate



Staged
Basic

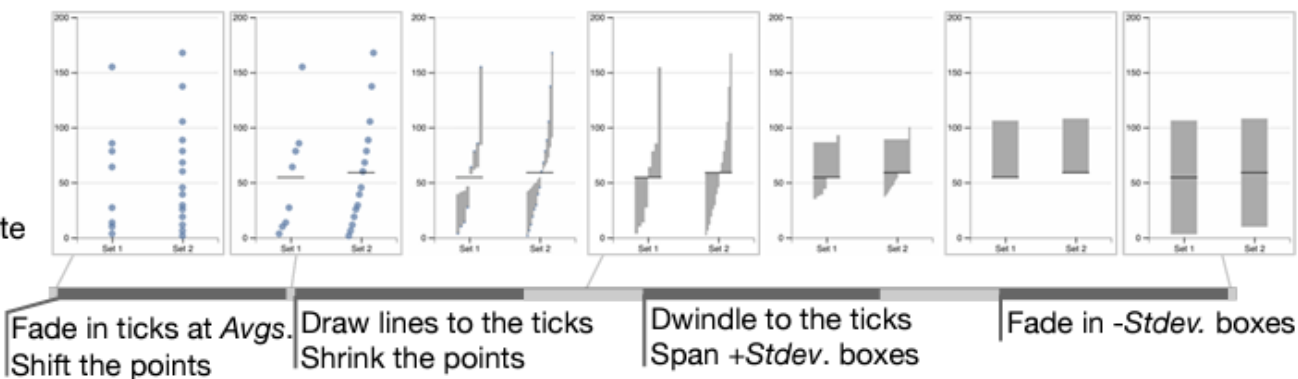


Median

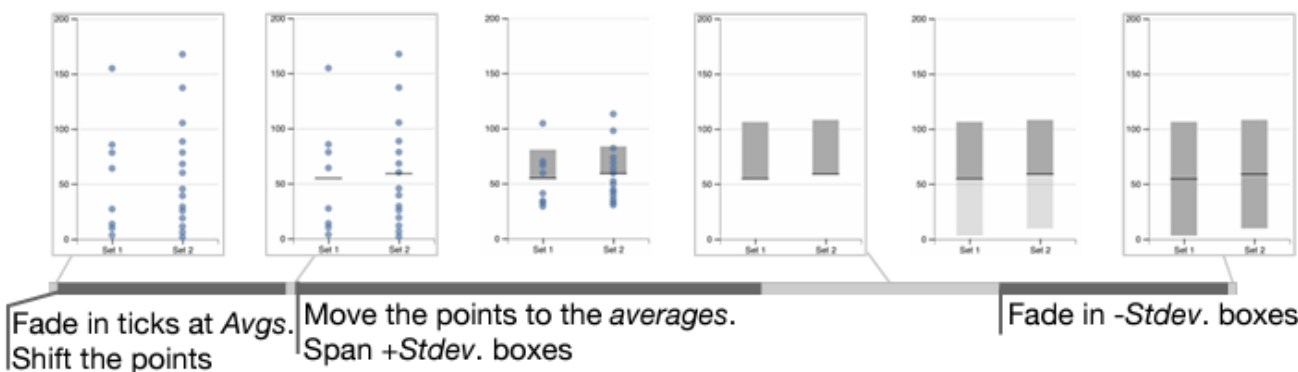


Standard Deviation

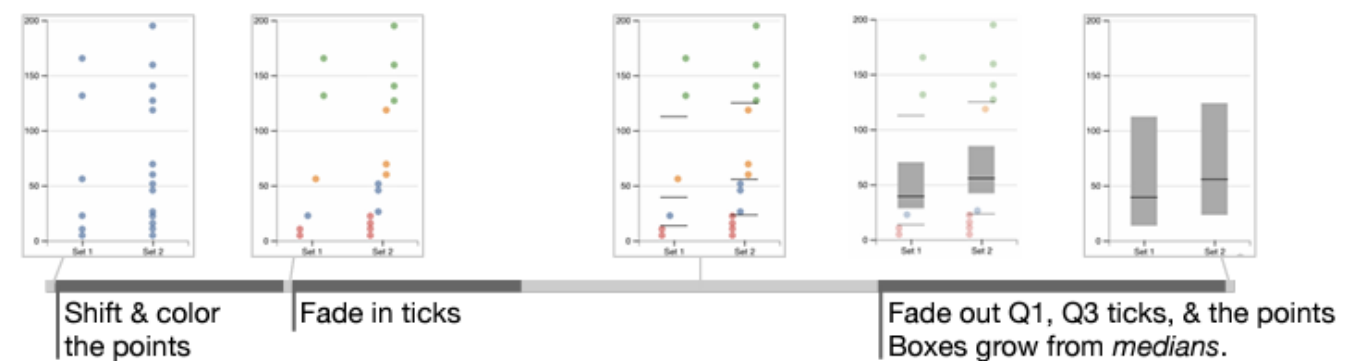
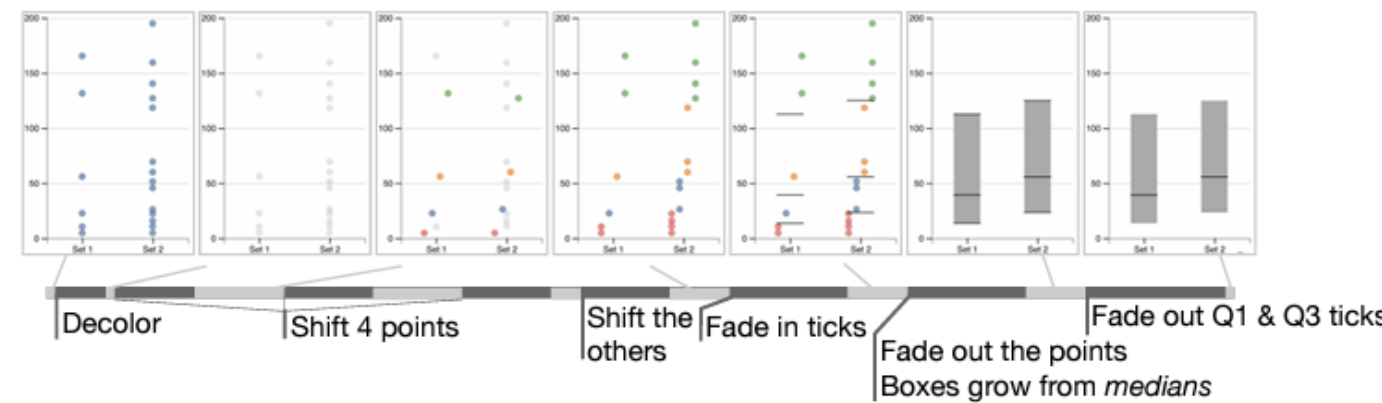
Staged
Elaborate



Staged
Basic



Interquartile Range



You try!

Consider an animated transition between two views (e.g., moving from a stacked bar chart to a grouped bar chart).

Sketch the beginning and end states.

Sketch possible keyframes between them.

Spend 5 minutes thinking about it, and then share with your neighbor.